



3D Transformation

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Motivation

■ Geometric transformations

- Translation, rotation, reflection
- Scaling, shearing
- Orthogonal projection, perspective projection

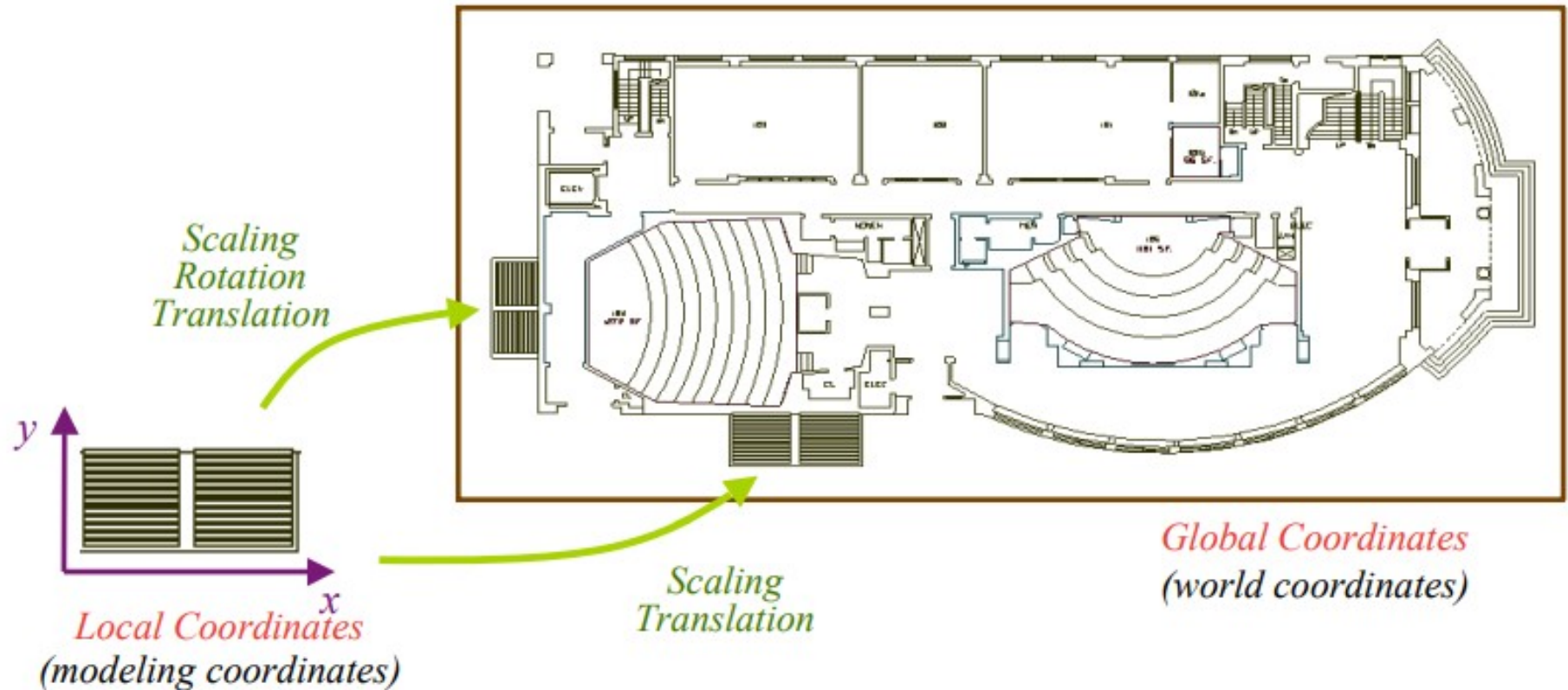
■ Why are geometric transformations necessary?

- for positioning geometric objects in 2D and 3D.
- for modelling geometric objects in 2D and 3D
- For viewing geometric objects in 2D and 3D.

Motivation (cont.): modelling objects in 2D

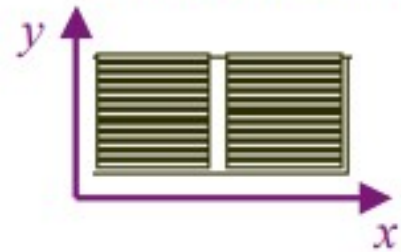
- Geometric transformations can specify object modelling operations
 - They allow us to define an object through its local coordinate system (modeling coordinates)
 - They allow us to define an object several times in a scene with a global coordinate system (world coordinates)

Motivation (cont.): modelling objects in 2D

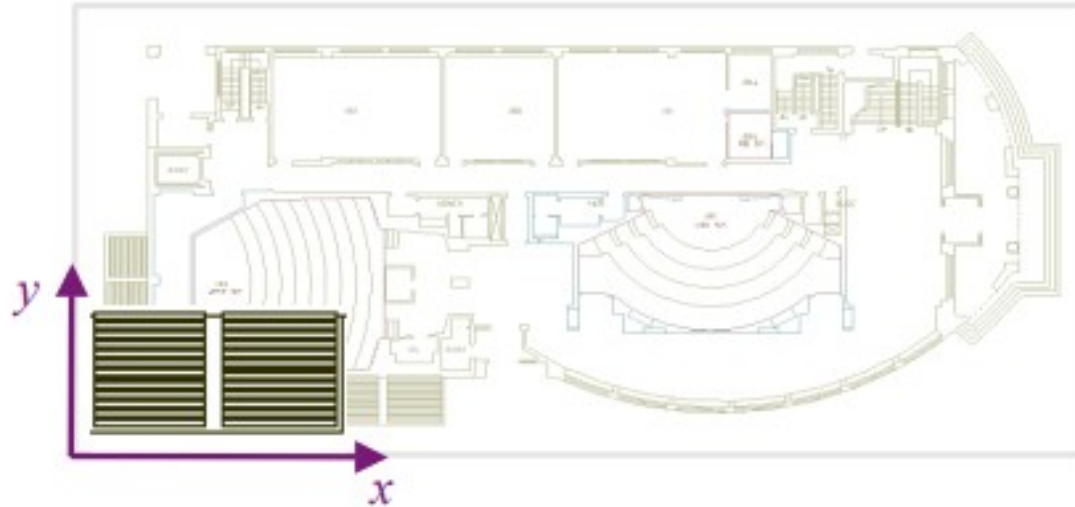


Motivation (cont.): modelling objects in 2D

Local Coordinates

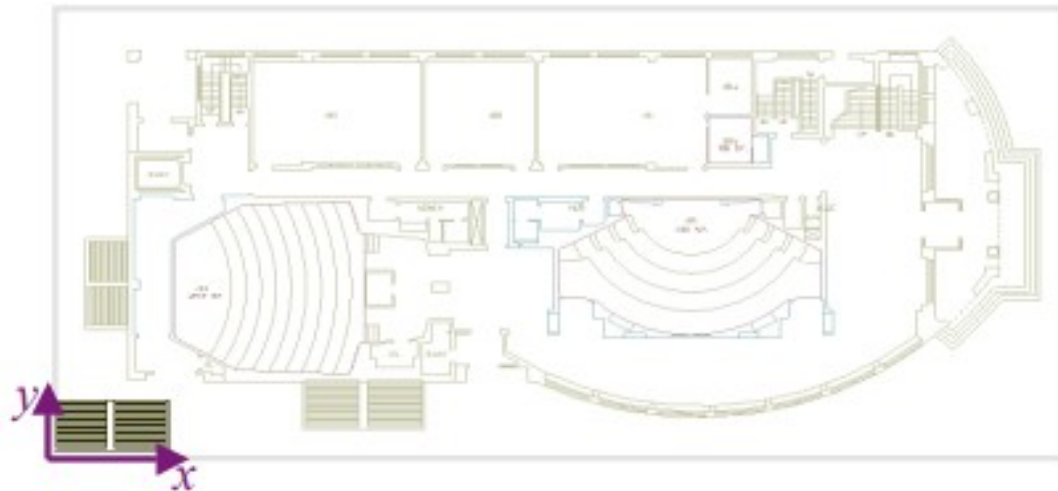


Positioning



Global Coordinates

Scaling



3D Point Representation

- A 3D point P is represented in homogeneous coordinates by a 4-dimensional vector:

$$P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- As for 2D points:

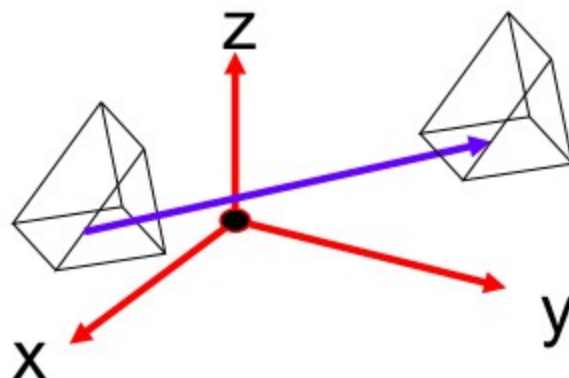
$$p = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \alpha x \\ \alpha y \\ \alpha z \\ \alpha \end{bmatrix}$$

3D Translation

P is translated to P' by:

$$\begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ z + t_z \\ 1 \end{bmatrix}$$

Or: $T \cdot P = P'$



Inverse: $T^{-1} \cdot P' = P$

3D Translation

- A position $P=(x, y, z)$ in 3D space is translated to a location $P'=(x', y', z')$ by adding translation distances t_x , t_y , and t_z :

$$\begin{aligned}x' &= x + t_x & y' &= y + t_y & z' &= z + t_z\end{aligned}\tag{9-1}$$

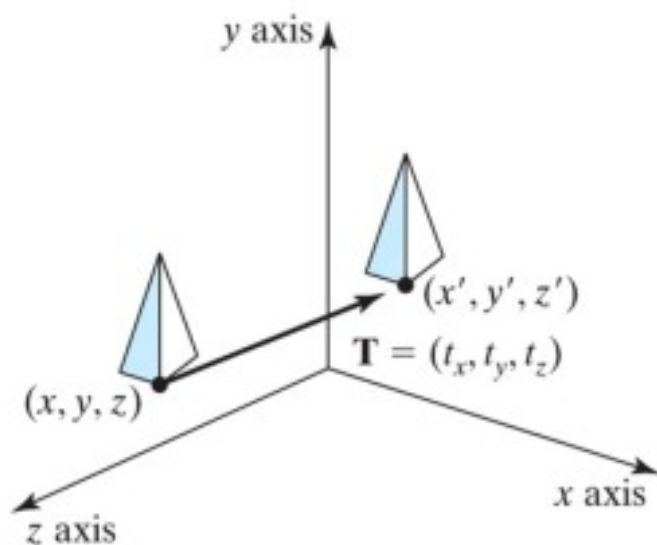


FIGURE 9-2 Shifting the position of a three-dimensional object using translation vector T .

By matrix form:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}\tag{9-2}$$

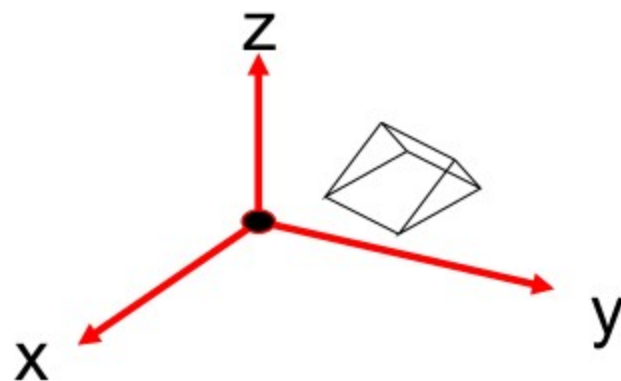
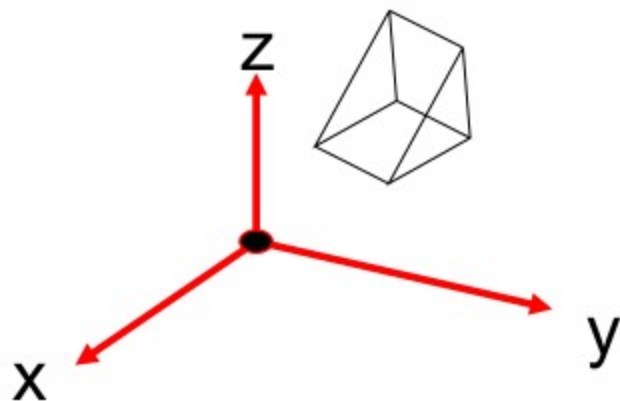
or

$$P' = T \cdot P\tag{9-3}$$

3D Scaling

$$\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} ax \\ by \\ cz \\ 1 \end{bmatrix}$$

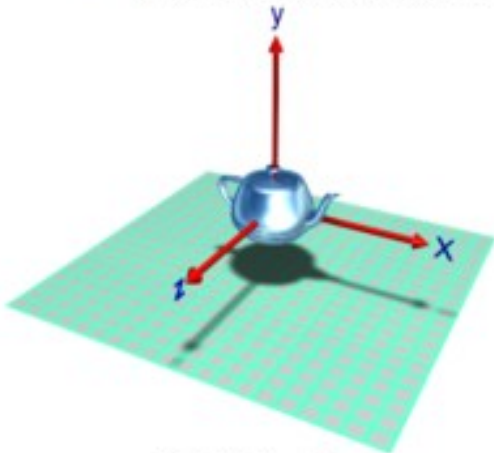
Or $S P = P'$



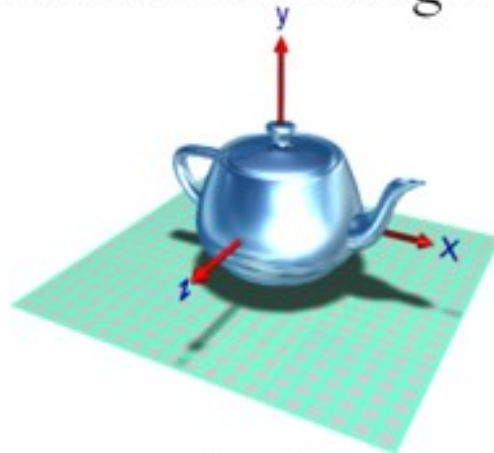
$$S^{-1} P' = P$$

3D Scaling

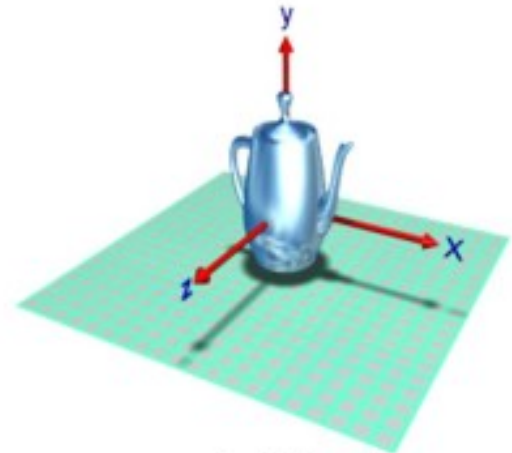
- Scale objects relative to the coordinate origin (0, 0, 0)
 - All vectors are scaled from the origin



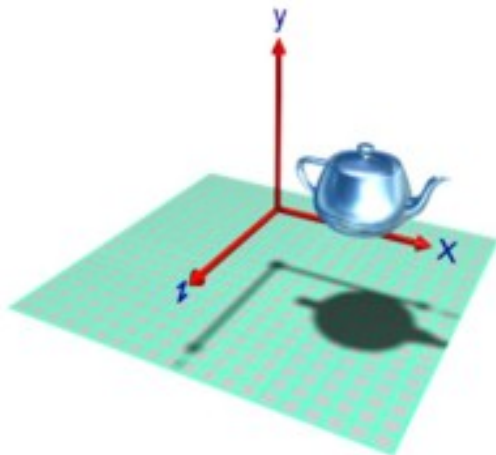
Original



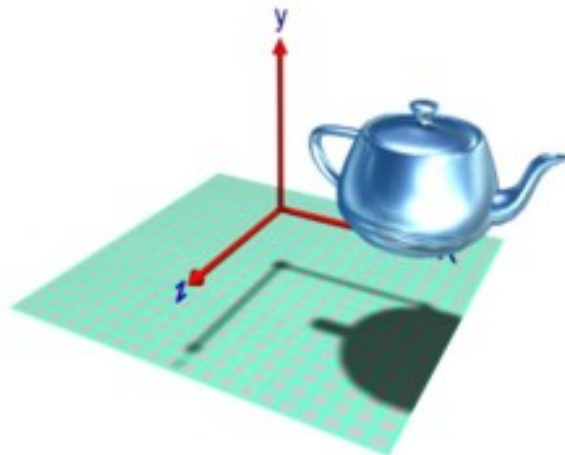
scale all axes



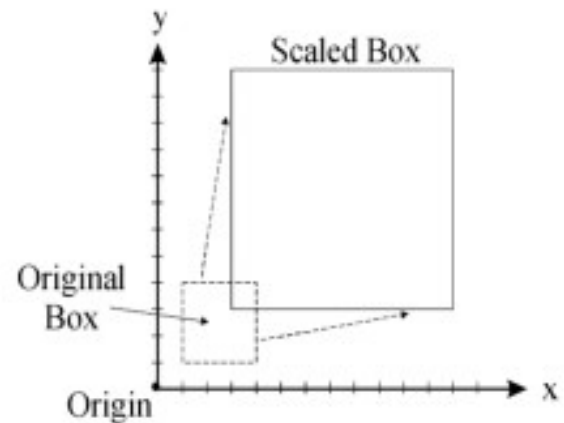
scale Y axis



offset from origin

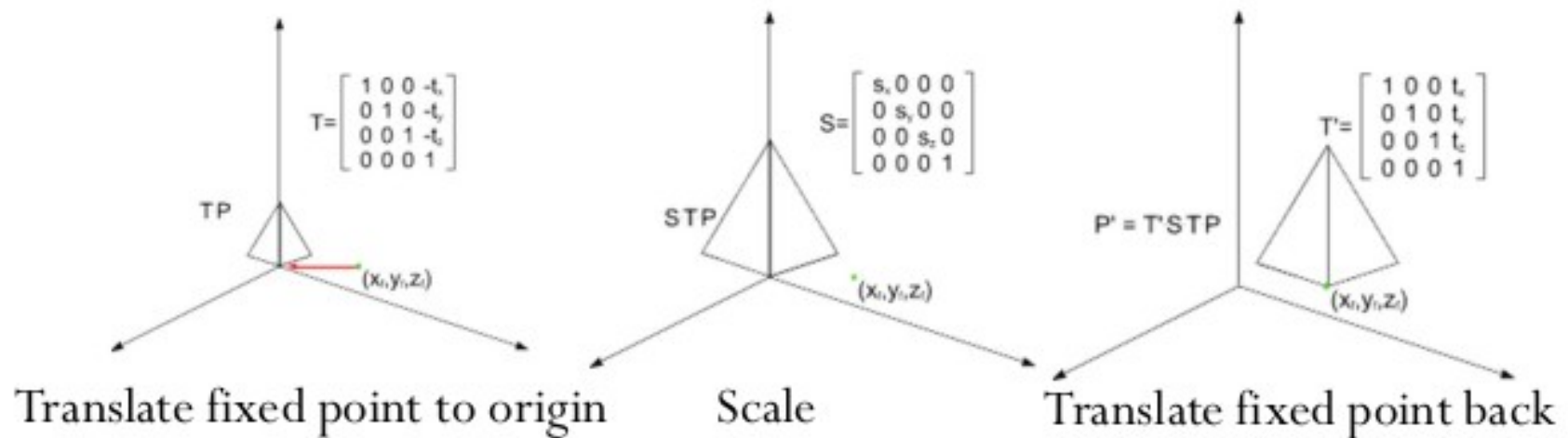
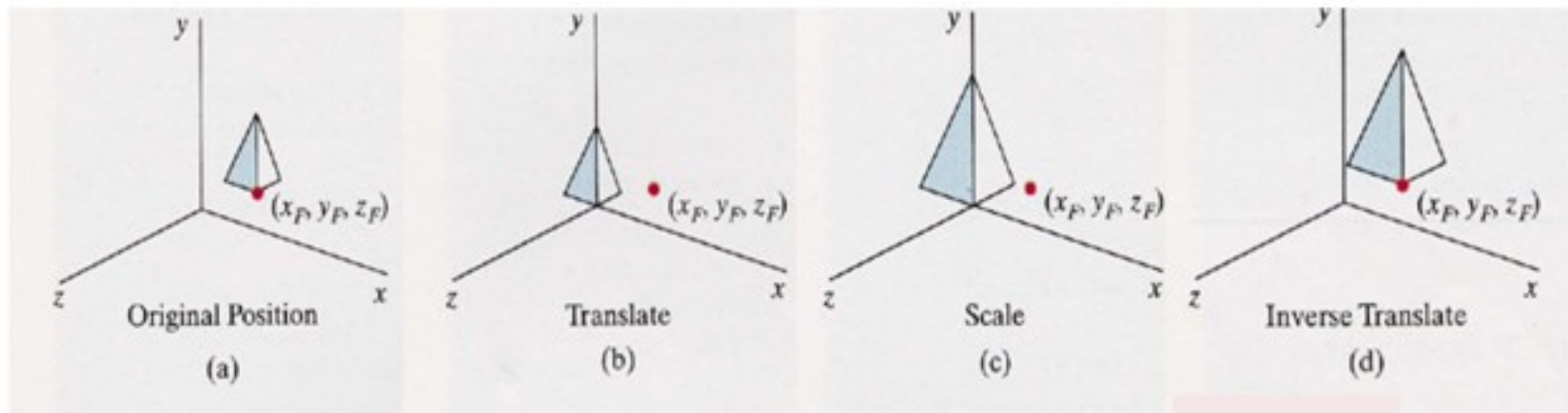


distance from origin also scales



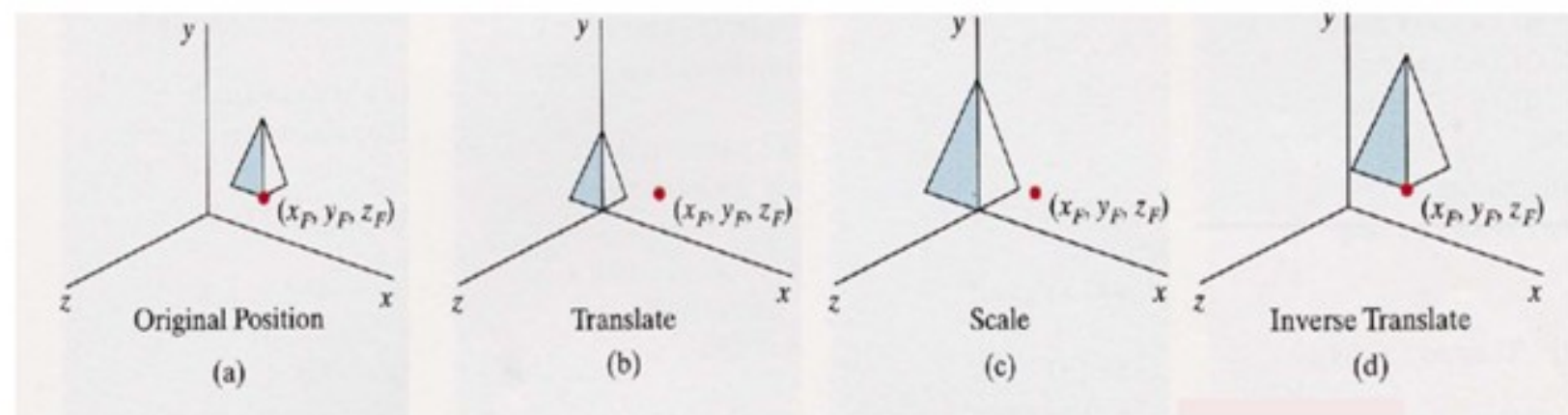
3D Scaling

- Scaling objects relative to a selected fixed point (x_f, y_f, z_f)



3D Scaling

- Scaling objects relative to a selected fixed point (x_f, y_f, z_f) (cont.)

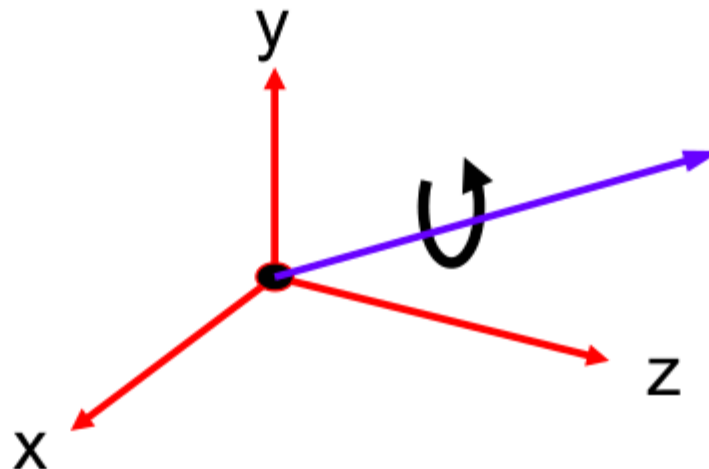


$$T(x_f, y_f, z_f) \cdot S(s_x, s_y, s_z) \cdot T(-x_f, -y_f, -z_f) = \begin{bmatrix} 1 & 0 & 0 & x_f \\ 0 & 1 & 0 & y_f \\ 0 & 0 & 1 & z_f \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -x_f \\ 0 & 1 & 0 & -y_f \\ 0 & 0 & 1 & -z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T(x_f, y_f, z_f) \cdot S(s_x, s_y, s_z) \cdot T(-x_f, -y_f, -z_f) = \begin{bmatrix} s_x & 0 & 0 & (1-s_x)x_f \\ 0 & s_y & 0 & (1-s_y)y_f \\ 0 & 0 & s_z & (1-s_z)z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Rotation

- To generate a rotation in 3D we have to specify:
 - axis of rotation (2 d.o.f.)
 - amount of rotation (1 d.o.f.)
- Note, the axis passes through the origin

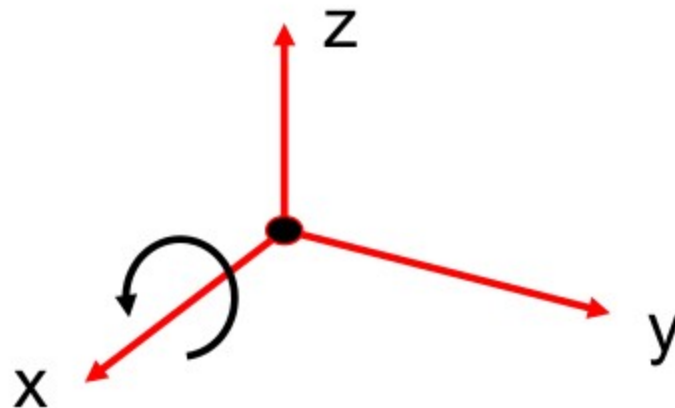


3D Rotation

- Counterclockwise rotation about x-axis

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$p' = R_x(\theta) p$$

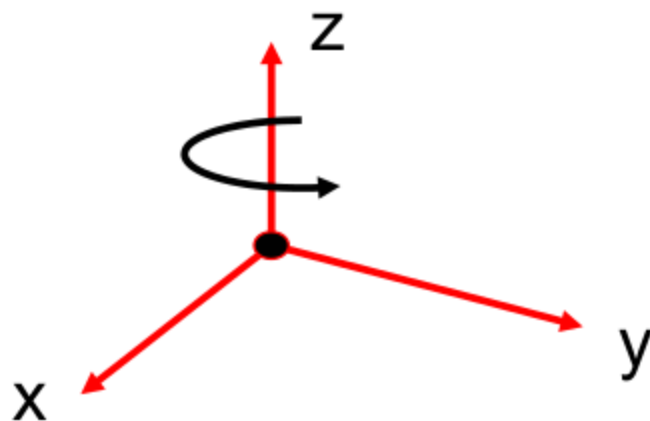


3D Rotation

- Counterclockwise rotation about z-axis

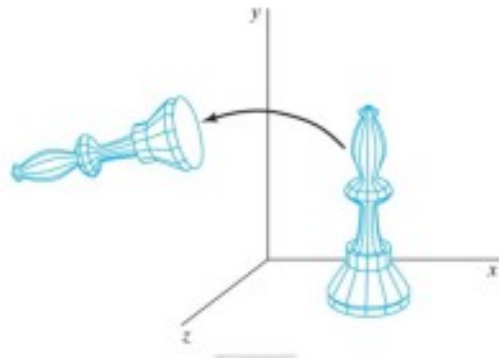
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$p' = R_z(\theta) p$$



3D Coordinate Axis Rotation

- Rotation of an object about the z axis



$$x' = x \cos \theta - y \sin \theta$$

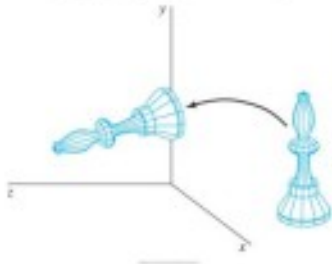
$$y' = x \sin \theta + y \cos \theta$$

$$z' = z$$

$$T = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_x = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

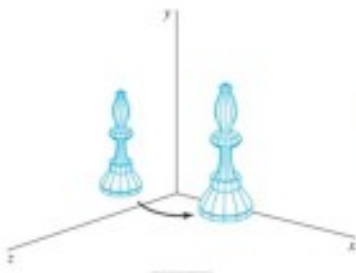
- Rotation of an object about the x axis



$$z \rightarrow x; x \rightarrow y; y \rightarrow z.$$

$$R_z = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Rotation of an object about the y axis

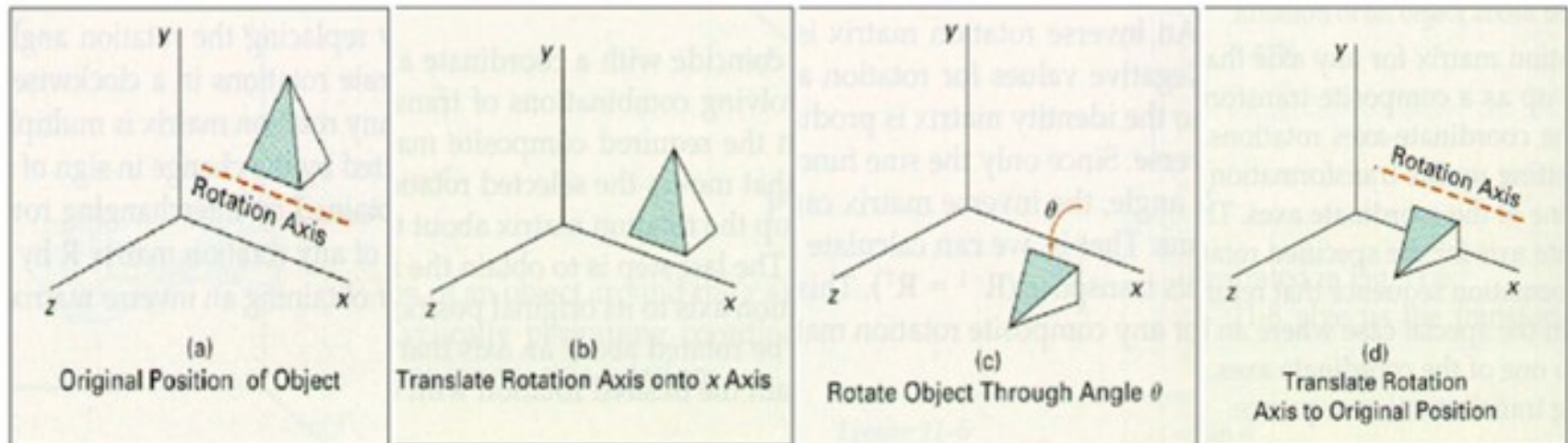


$$x \rightarrow y; y \rightarrow z; z \rightarrow x.$$

$$R_y = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

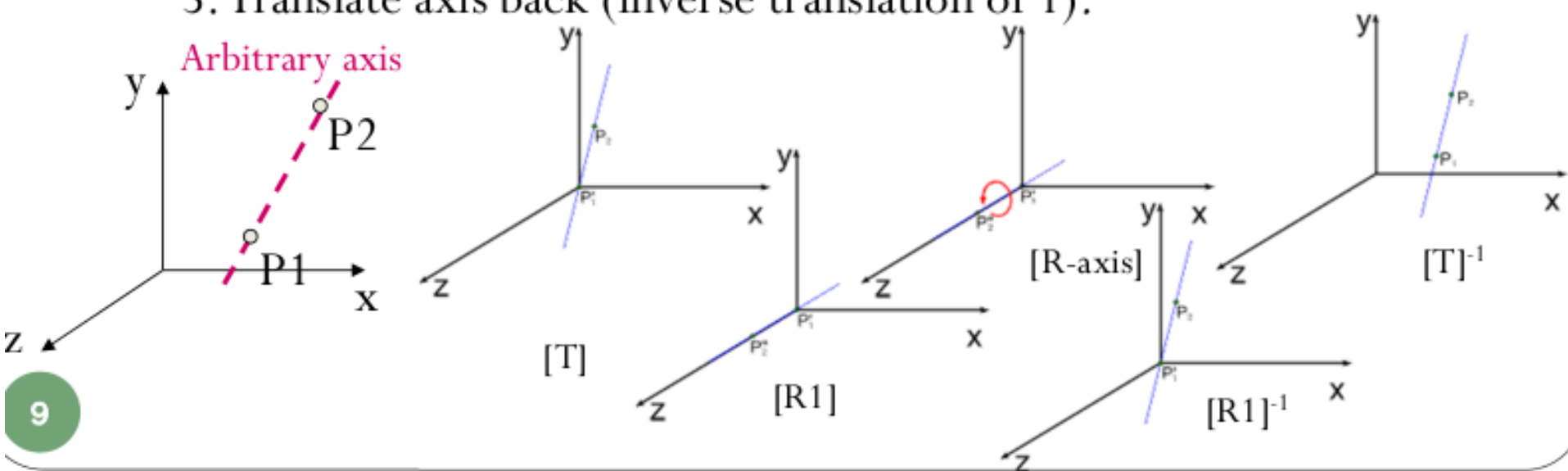
3D Rotations Parallel to Axes

- Rotation an axis that is parallel to one of the coordinate axes
 - Translate the object so that the rotation axis coincides with the parallel coordinate axis
 - Perform the specified rotation about that axis
 - Translate the object so that the rotation axis is moved back to its original position

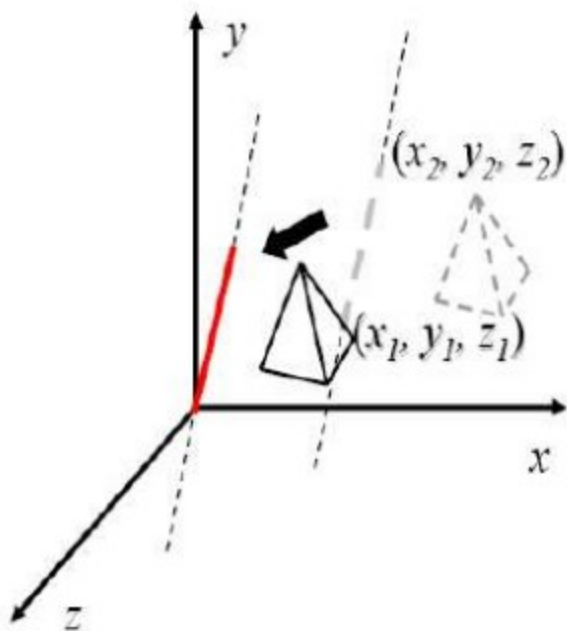


3D Rotations about Arbitrary Axis

- Rotate about the arbitrary axis through P1 and P2:
 1. Translate P1 to origin.
 2. Rotate so that the rotation axis is aligned with one of the principle coordinate axes.
 3. Perform the desired rotation about coordinate axis.
 4. Rotate axis back (inverse rotation of 2).
 5. Translate axis back (inverse translation of 1).

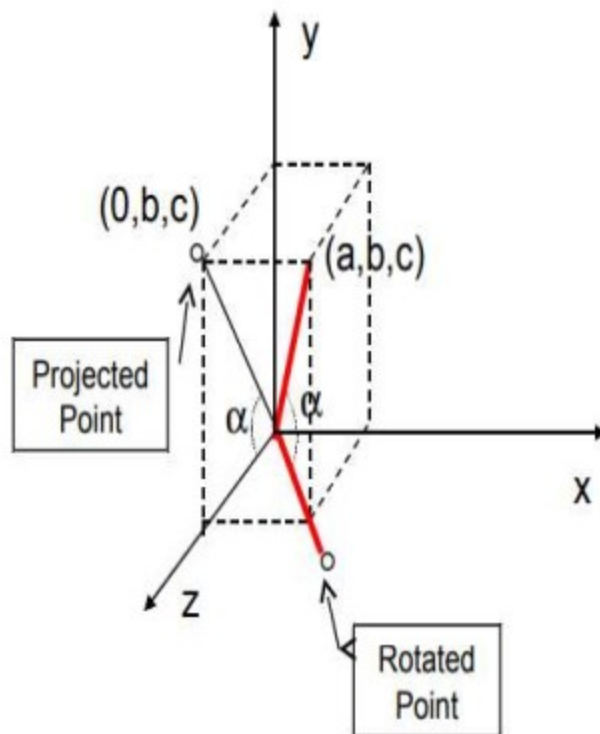


- Step 1. Translate



$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 2. Rotate about x axis by α

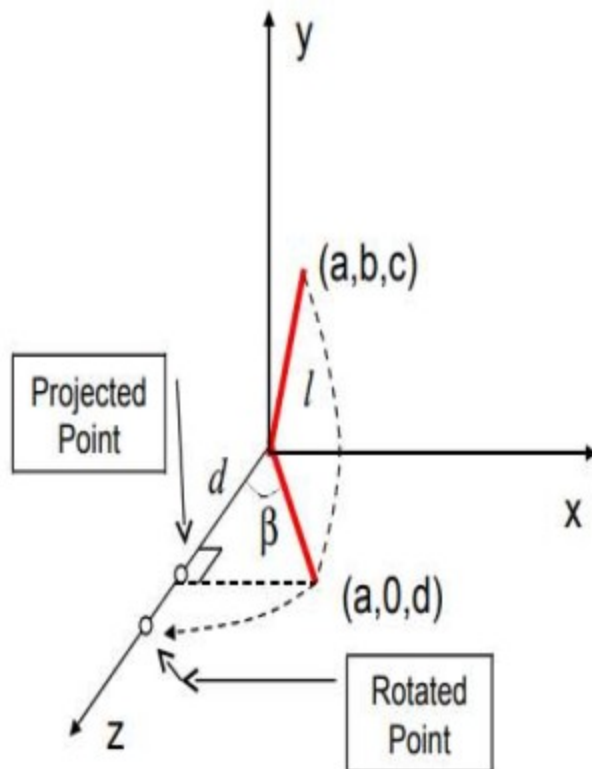


$$\sin \alpha = \frac{b}{\sqrt{b^2 + c^2}} = \frac{b}{d}$$

$$\cos \alpha = \frac{c}{\sqrt{b^2 + c^2}} = \frac{c}{d}$$

$$\mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c/d & -b/d & 0 \\ 0 & b/d & c/d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 3. Rotate about y axis by β (clockwise)



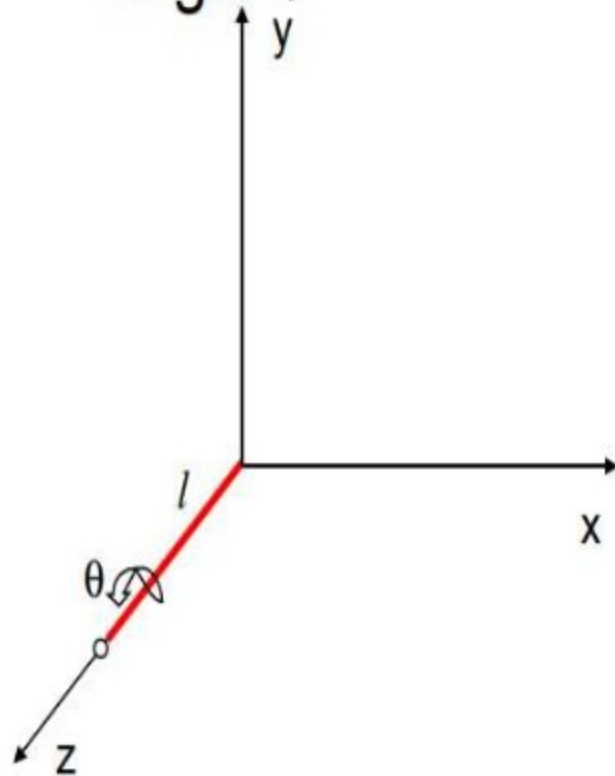
$$\sin \beta = \frac{a}{l}, \quad \cos \beta = \frac{d}{l}$$

$$l^2 = a^2 + b^2 + c^2 = a^2 + d^2$$

$$d = \sqrt{b^2 + c^2}$$

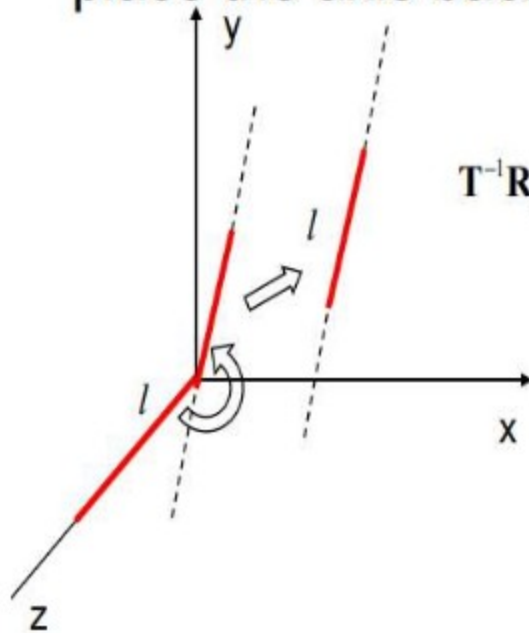
$$\mathbf{R}_y(\beta) = \begin{bmatrix} \cos \beta & 0 & -\sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ \sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} d/l & 0 & -a/l & 0 \\ 0 & 1 & 0 & 0 \\ a/l & 0 & d/l & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Step 4. Rotate about z axis by the desired angle θ



$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Step 5. Apply the reverse transformation to place the axis back in its initial position



$$\mathbf{T}^{-1}\mathbf{R}_x^{-1}(\alpha)\mathbf{R}_y^{-1}(\beta) = \begin{bmatrix} 1 & 0 & 0 & -x_l \\ 0 & 1 & 0 & -y_l \\ 0 & 0 & 1 & -z_l \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha & 0 \\ 0 & -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos \beta & 0 & \sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}(\theta) = \mathbf{T}^{-1}\mathbf{R}_x^{-1}(\alpha)\mathbf{R}_y^{-1}(\beta)\mathbf{R}_z(\theta)\mathbf{R}_y(\beta)\mathbf{R}_x(\alpha)\mathbf{T}$$

Matrix Composition

- Transformations can be combined by matrix multiplication

$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \left(\begin{bmatrix} 1 & 0 & 0 & tx \\ 0 & 1 & 0 & ty \\ 0 & 0 & 1 & tz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} sx & 0 & 0 & 0 \\ 0 & sy & 0 & 0 \\ 0 & 0 & sz & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

$\mathbf{p}' = \quad T(t_x, t_y) \quad R(Q) \quad S(s_x, s_y) \quad \mathbf{p}$

$$\mathbf{p}' = (T * (R * (S * \mathbf{p})))$$

$$\mathbf{p}' = (T * R * S) * \mathbf{p}$$

- Order of transformations**

- Matrix multiplication is not commutative

$$\mathbf{p}' = T * R * S * \mathbf{p}$$

$\longleftrightarrow$
 “Global” “Local”

- **Questions:** Translate the 3D square with vertices $A(4,5,7)$, $B(-3,-2,1)$, $C(2,4,7)$ and $D(-5,3,2)$ using translation vector $(1,3,4)$.

- **Question1:** A 3D object with vertices $A(-2,-4,4)$, $B(3,-6,-8)$, $C(-6,1,0)$ and $D(3,-6,2)$ is required to scale with scale factor $S(2,4,6)$ about origin. Find the final coordinates.
- **Question2:** Given a solid PQT, $p(2,-5,0)$, $Q(4,-2,3)$ and $T(6,-7,-1)$ is translated with $T(9,0,-2)$ and then scale about origin with scaling factor $(2,4,5)$. Find the final coordinates.
- **Question3:** suppose we have a quadrilateral ABCD with vertices $A(3,1,4)$, $B(-7,-4,2)$, $C(5,-8,-3)$ and $D(3,0,1)$. What is the coordinates of quadrilateral if it is to be doubled in size about origin. Then translate obtained coordinates with $(-4,3,1)$
- **Question 4:** scale the triangle with vertices $A(-3,-2,1)$, $B(4,5,6)$, $C(5,7,9)$ to double its size about fixed point $(1,2,3)$.

- **Question1:** Rotate triangle with vertices $A()$, $B(0,2,1)$, $(2,3,0)$ and $(1,2,1)$ about the line joining the point $(0,0,0)$ and $(1,1,1)$ with the angle of rotation 45° in counter clockwise direction.
- **Question1:** Rotate triangle with vertices $A(0,2,1)$, $B(2,3,0)$ and $C(1,2,1)$ about the line joining the point $(2,2,2)$ and $(1,1,1)$ with the angle of rotation 45° in counter clockwise direction.



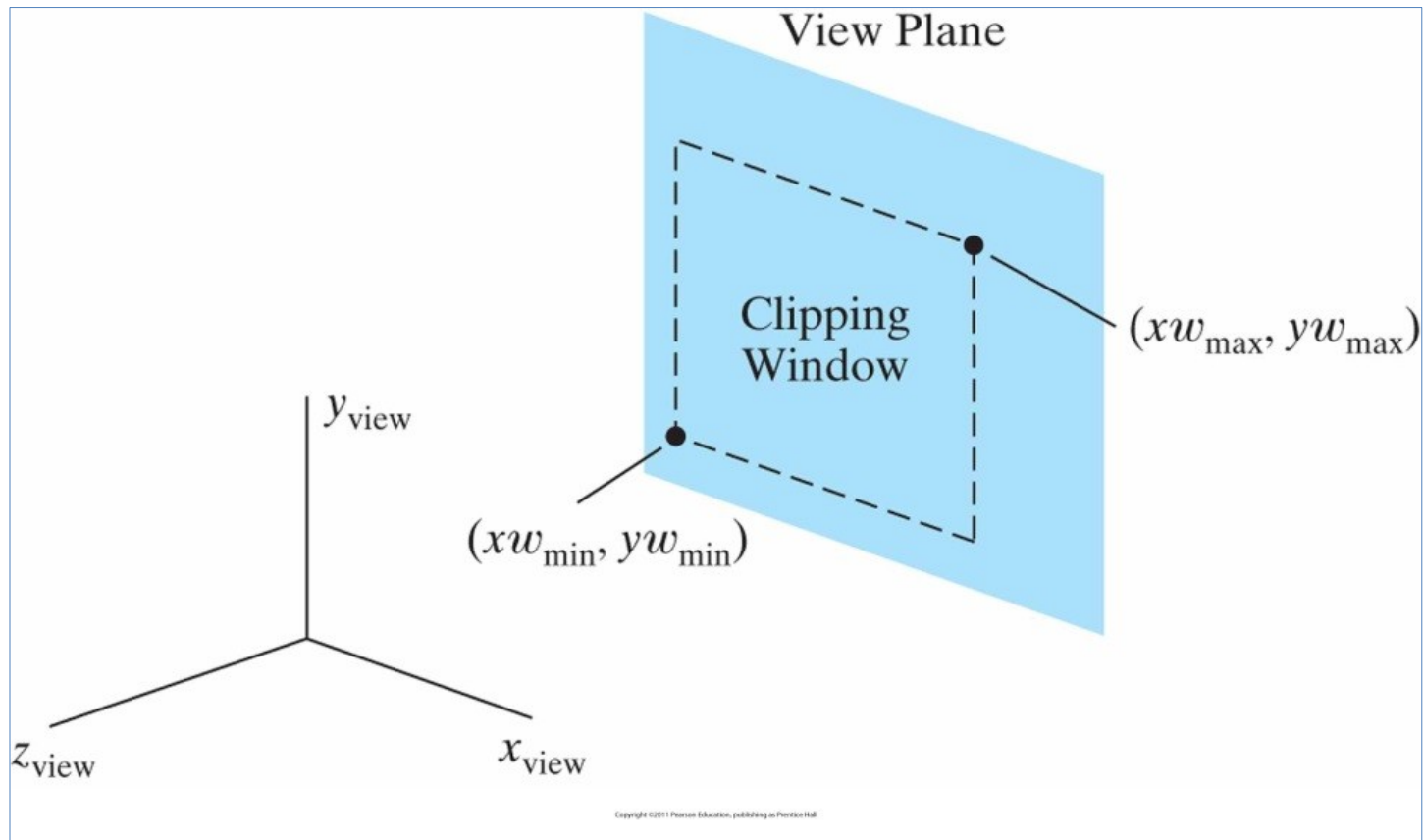
Projection Transformations

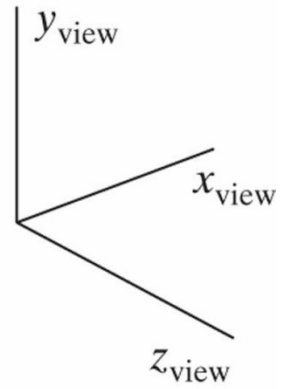
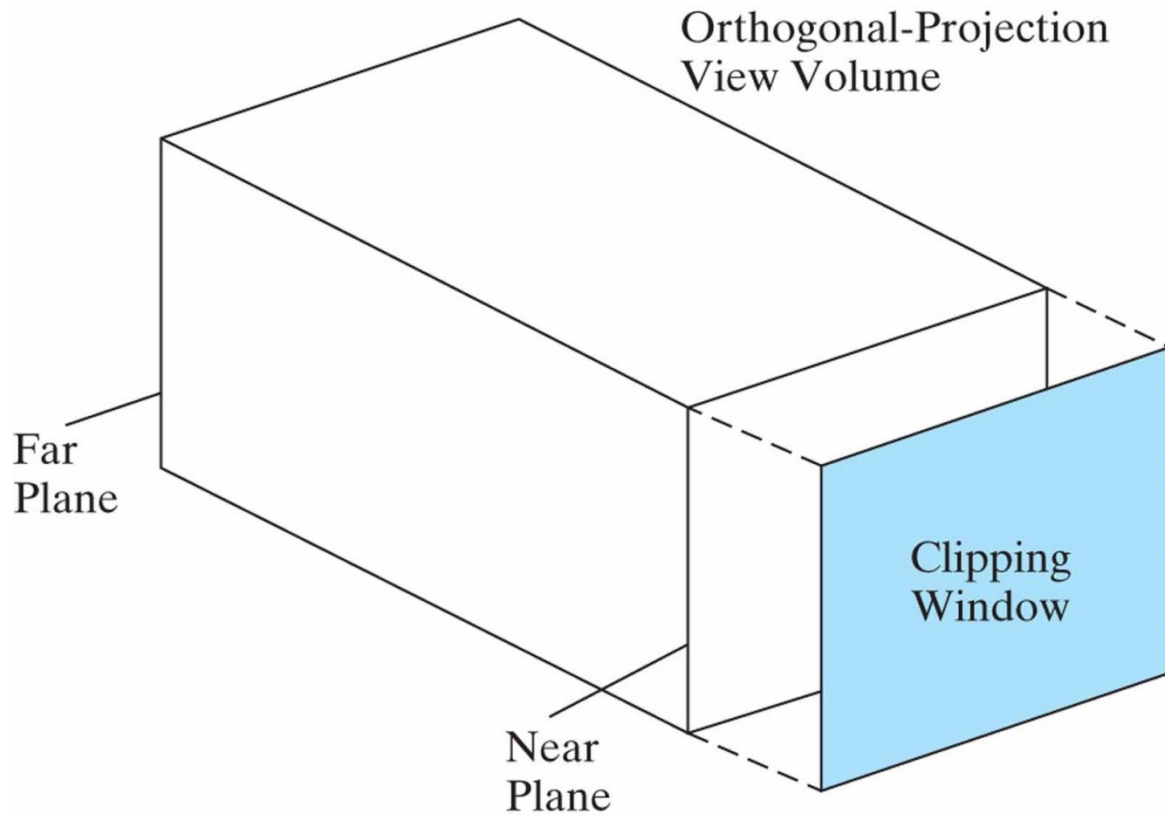
- As we knew that scene in view coordinate system is 3D.
- We need to display the scene on computer screen.
- Computer screen is 2D.
- So we need a way to transform 3D scene to 2D scene.
- This is called projection transformation.
- Projection transforms n dimension to $n-1$ dimension.
- We can project 2D point to 1D point
- So projection reduces dimension.



Projection Transformations

- We do not project all the scene in view coordinate or world coordinate.
- We define clipping window of some dimension.
- Then define view volume according to the size of clipping window.
- For this we define near plane and far plane.
- All the objects completely inside the view volume are projected on to the view plane.
- Others are clipped.







Projection Transformations

- For projection, we extend straight lines from each coordinates of objects inside view volume to view plane
- The intersection of these lines with view plane generates the projected object.

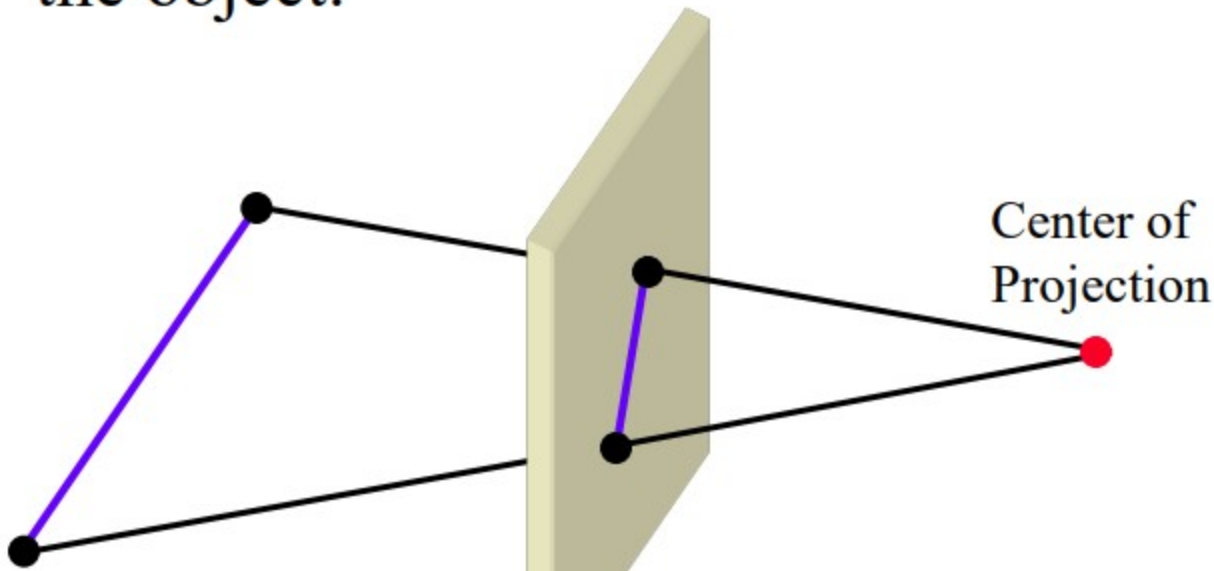


Projection Transformations

- Lines connecting projected point and actual point is called projectors.
- The projected point is collinear with the actual point on object and center of projection.
- Center of projection(COP) is the point where projectors meet.
- COP can be also taken as the location of camera or observer.
- The object is projected on a plane called view plane.

Projection Transformations

- A projection is formed by the intersection of certain lines (*projectors*) with the view plane.
- Projectors are lines from the *center of projection* through each point in the object.

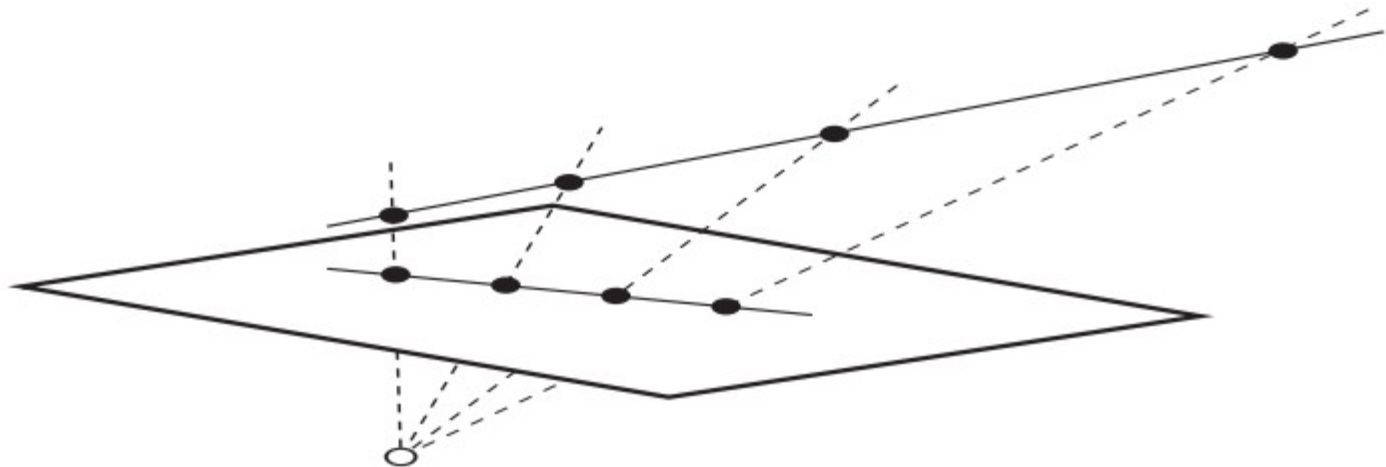


Properties of projections

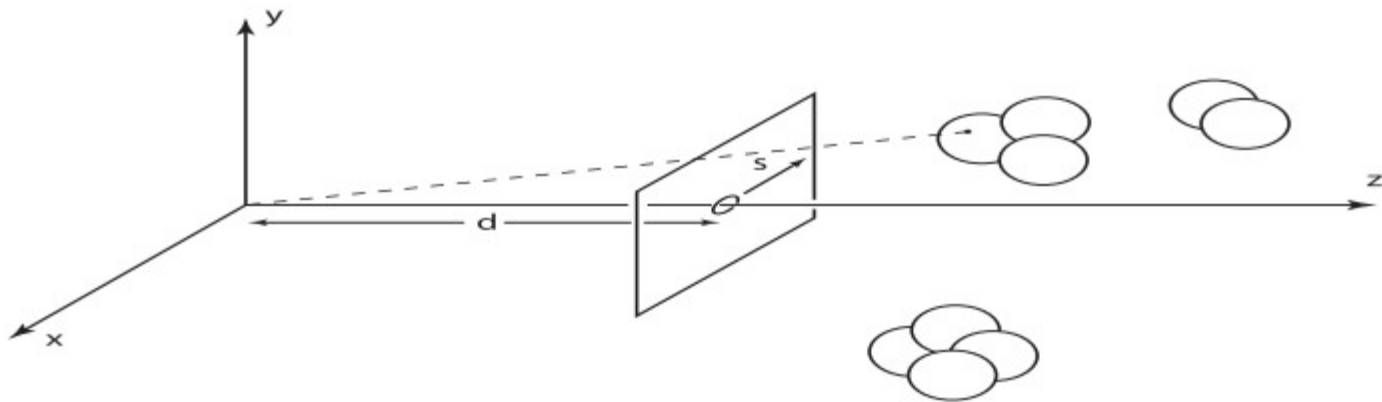
The perspective projection is an example of a **projective transformation**.

Here are some properties of projective transformations:

- ♦ Lines map to lines
- ♦ Parallel lines *don't* necessarily remain parallel
- ♦ Ratios are *not* preserved

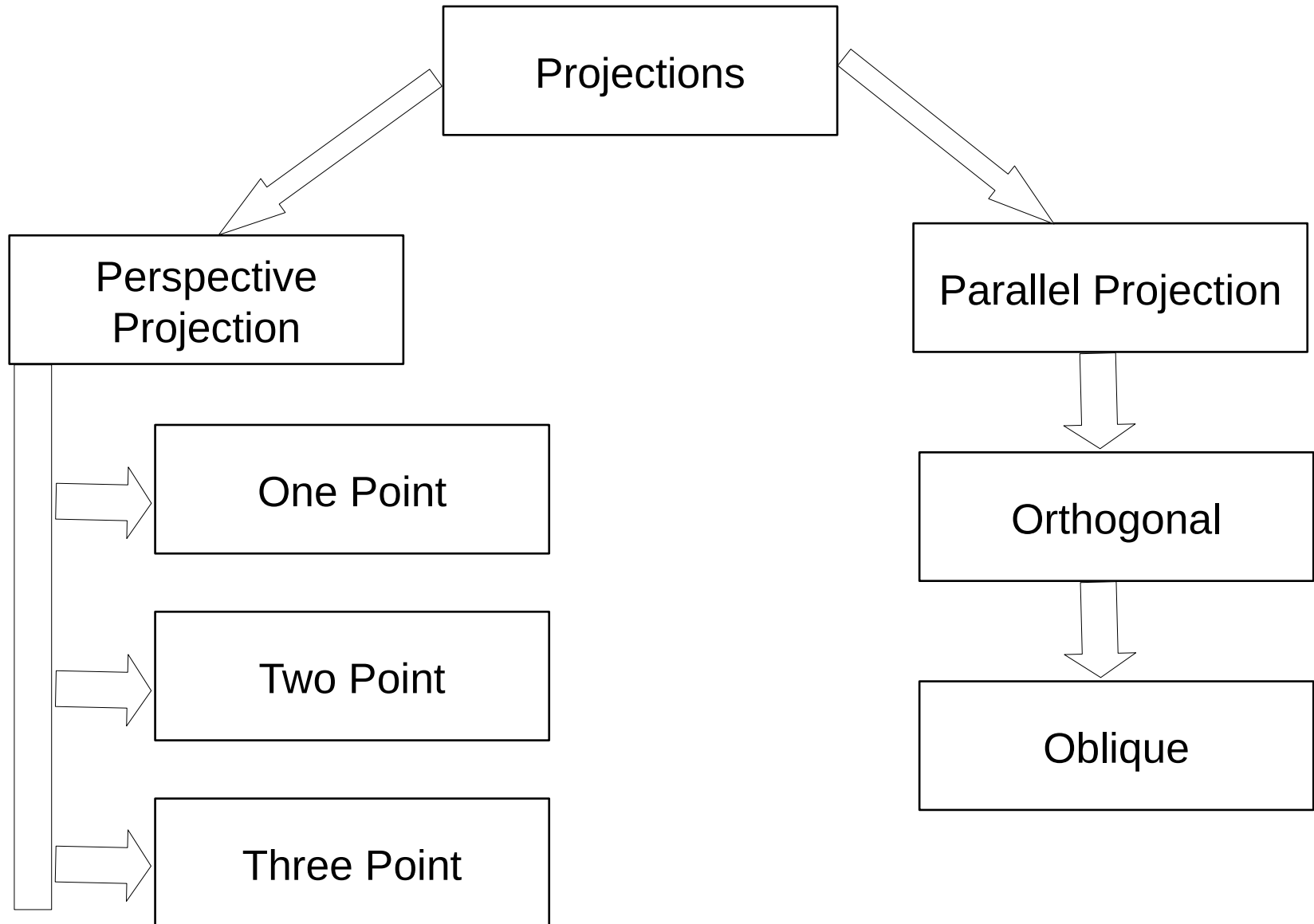


A typical eye space



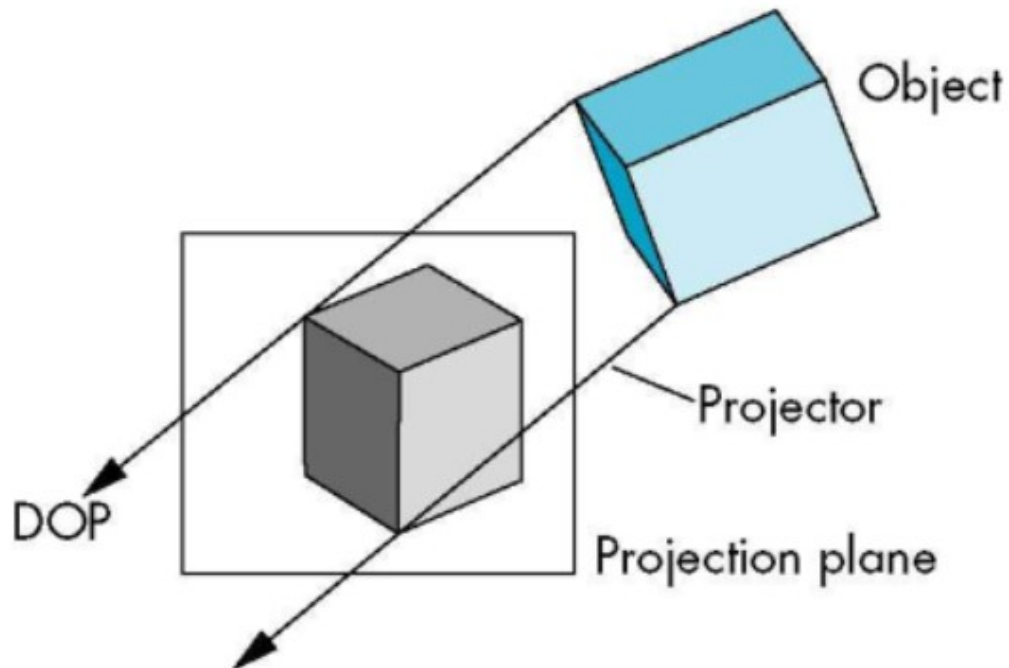
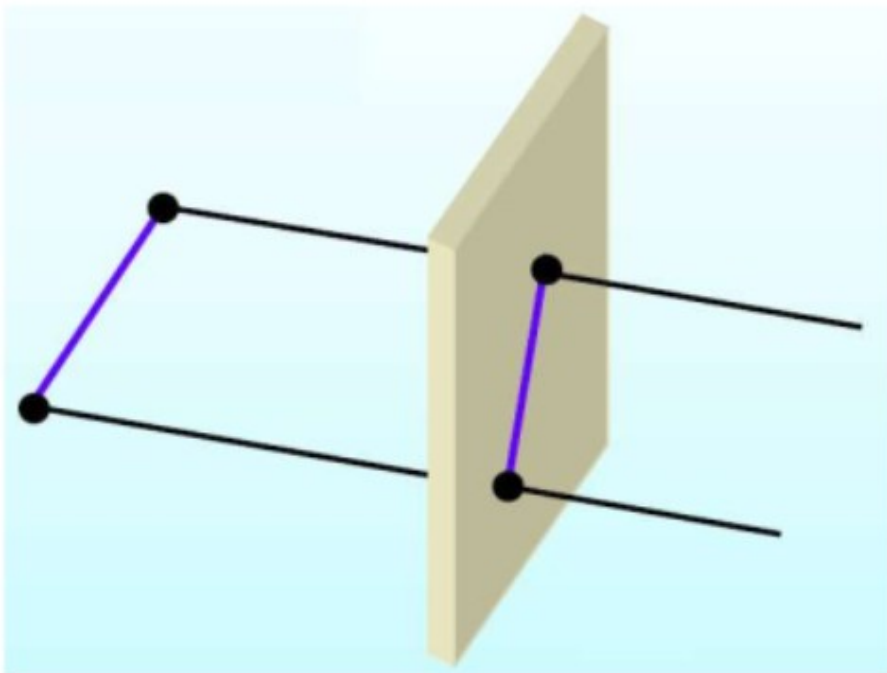
- ♦ **Eye**
 - Acts as the COP
 - Placed at the origin
 - Looks down the z -axis
- ♦ **Screen**
 - Lies in the PP
 - Perpendicular to z -axis
 - At distance d from the eye
 - Centered on z -axis, with radius s

Projection Types



Projection Transformations

- Projectors are parallel
- COP is at infinity.
- Parallel lines (Projectors) from an object are extended to the plane of projection to get projected object.

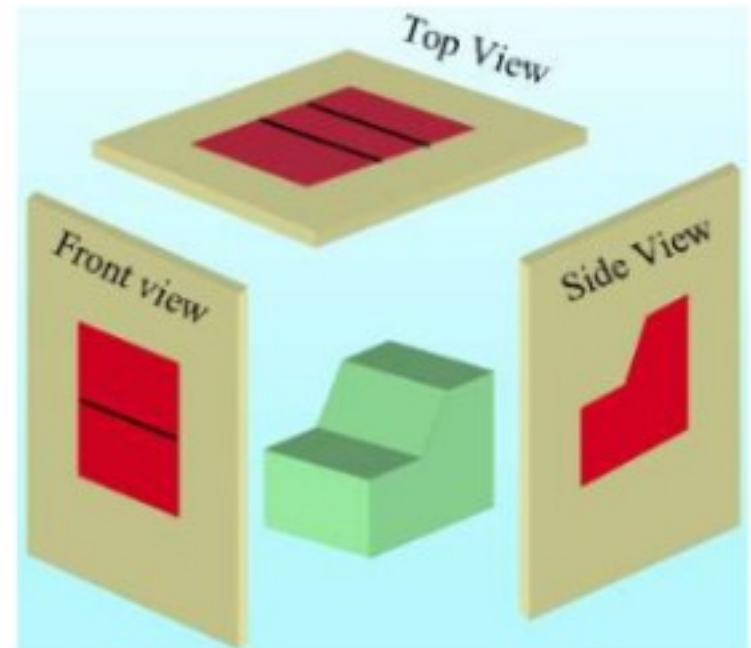
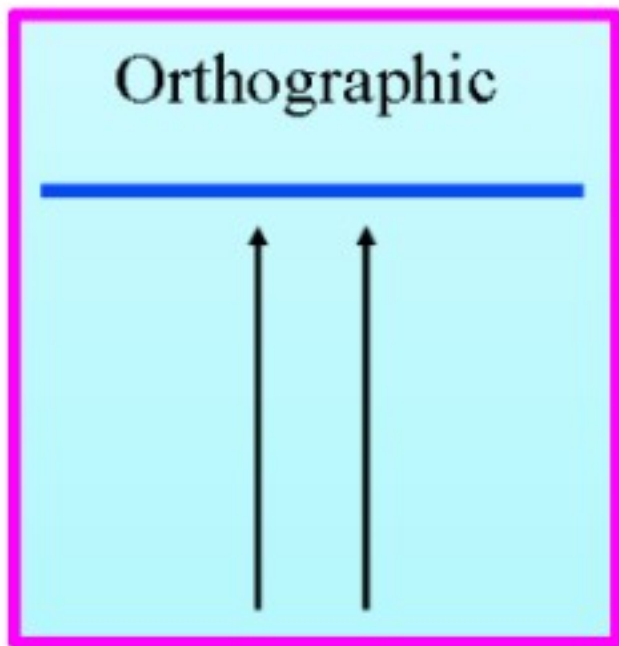


Parallel Projection

- Projectors are all parallel.
- **Orthographic**: Projectors are perpendicular to the projection plane.
- **Oblique**: Projectors are not necessarily perpendicular to the projection plane.

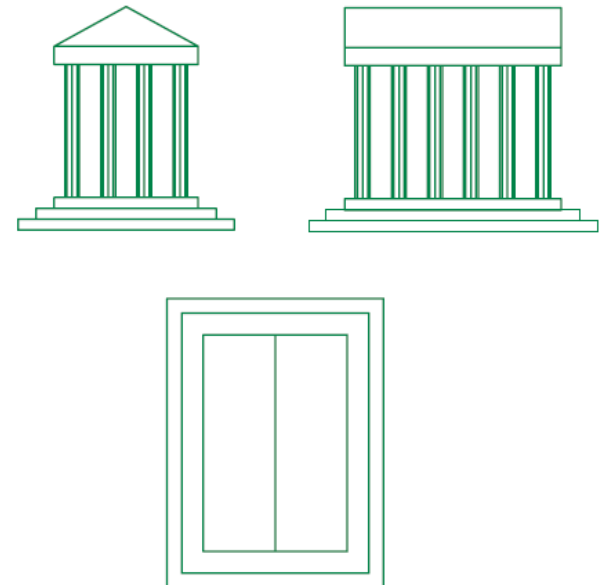
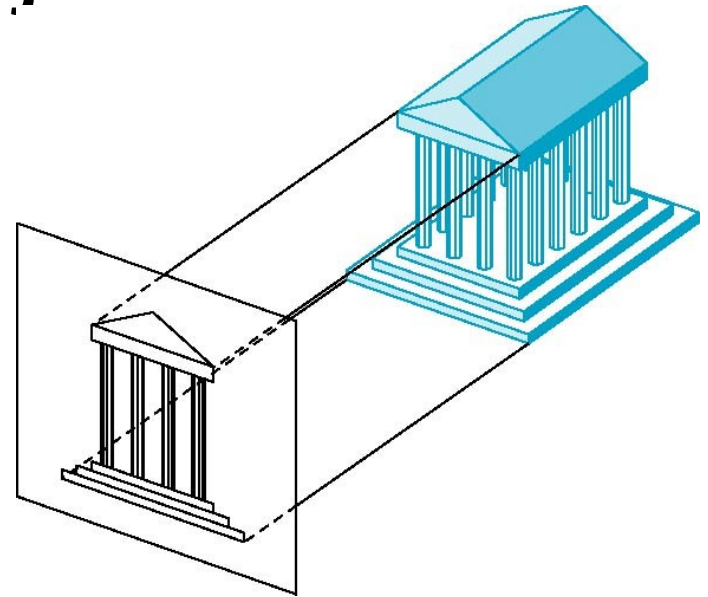
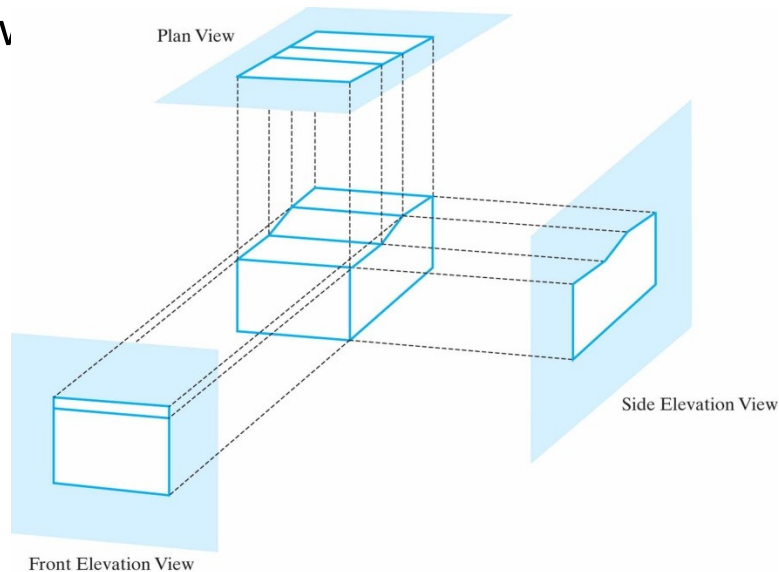
Orthographic Projection

- Projectors are perpendicular to plane of projection.
- Preserves shape and size.
- So, for 3D effect, top view, front view, and side view projections are necessary.



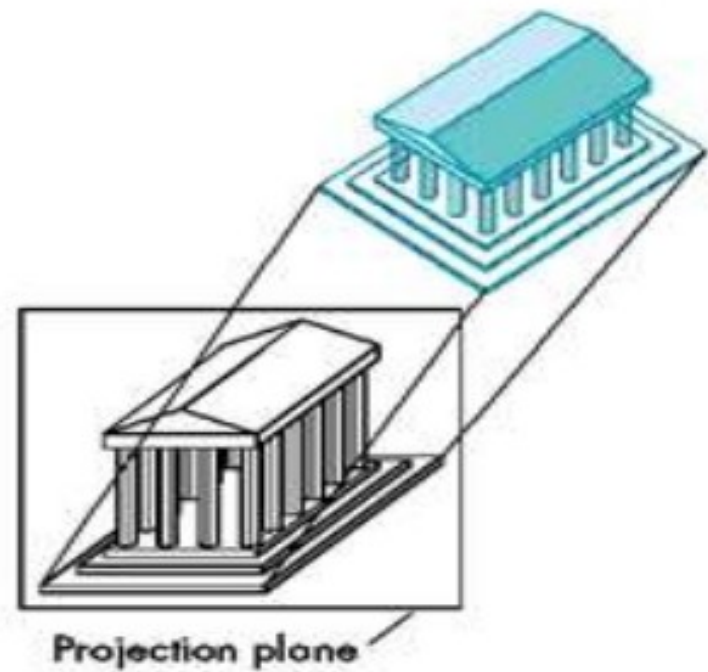
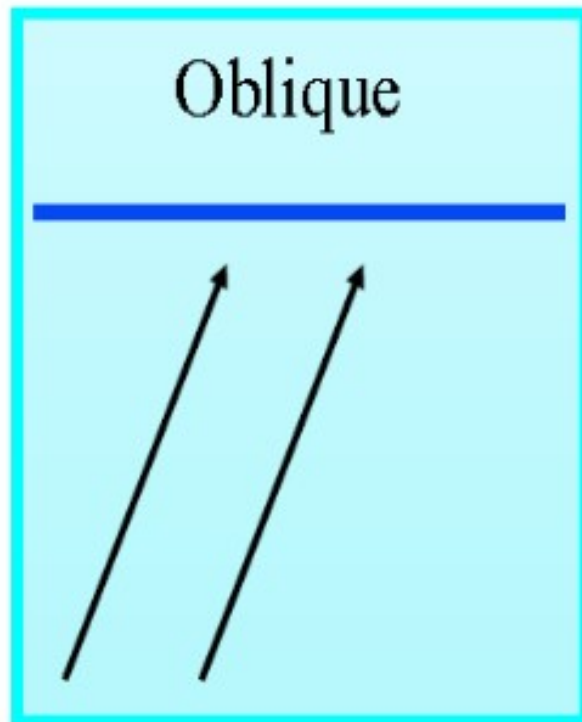
Orthogonal Projections

- A type of parallel projection in which direction of projection is perpendicular to the projection plane
 - Elevations (front, side), plan view (top)
- This type of view preserves distances and angles (for planes parallel to view)



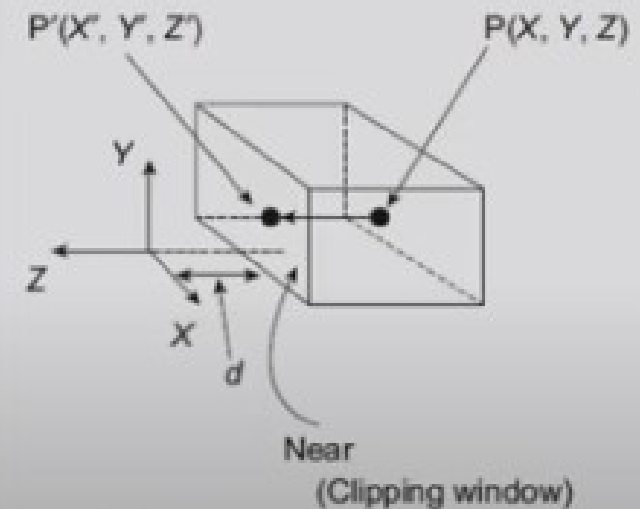
- **Oblique projection:**

- The projectors in oblique projection intersect the projection plane at an oblique angle(not perpendicular) to produce the projected image.



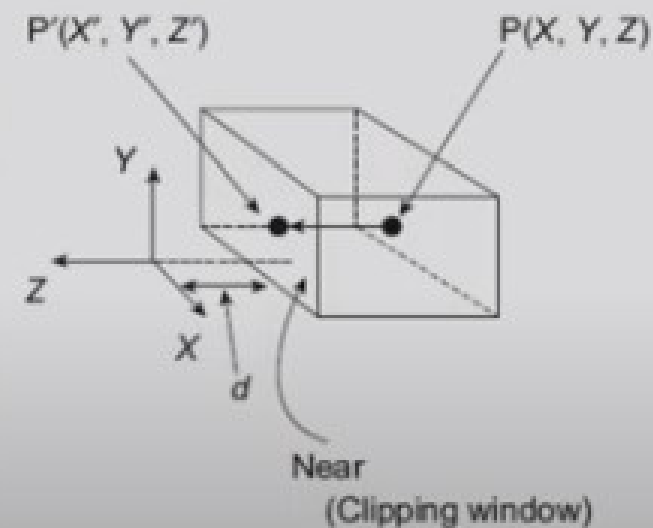
Transformation Matrix – Parallel Projection

- $\mathbf{P}$ is in view volume with coordinate (x, y, z) - projected as $\mathbf{P}'(x', y', z')$ on clipping window



Transformation Matrix – Parallel Projection

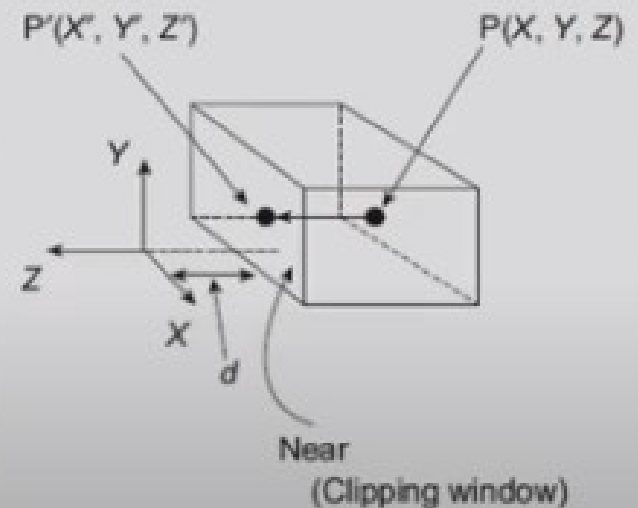
- Assuming near plane at a distance d along the $-z$ direction, coordinate of P' is
 - $x' = x$
 - $y' = y$
 - $z' = -d$



Transformation Matrix – Parallel Projection

- Transformation matrix for parallel projection

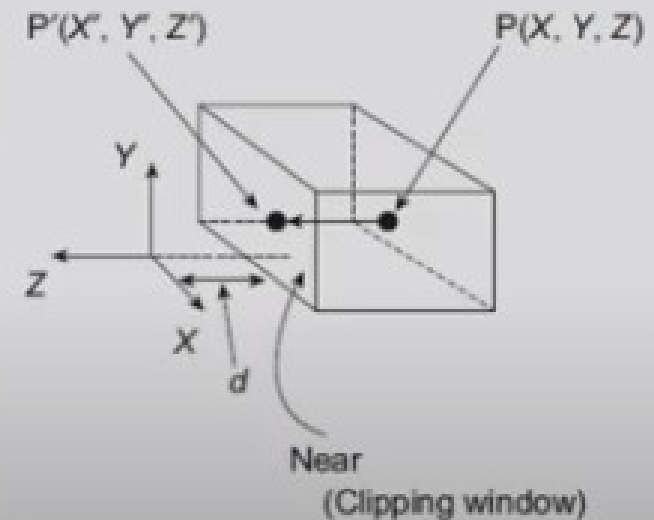
$$T_{par} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Transformation Matrix – Parallel Projection

- In terms of matrix multiplication we can write

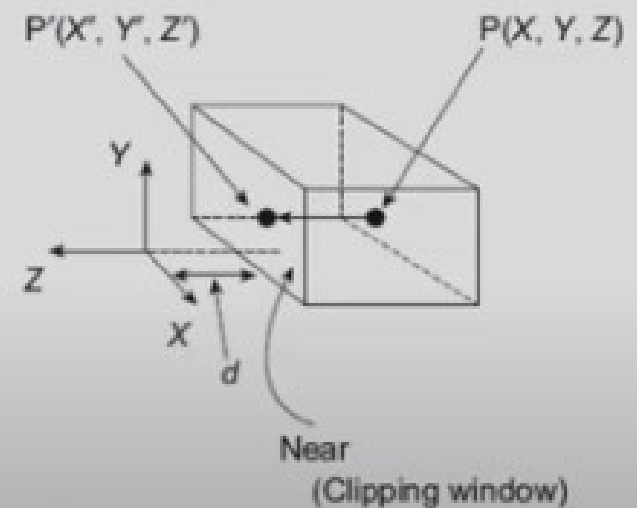
$$P' \begin{bmatrix} x'' \\ y'' \\ z'' \\ w \end{bmatrix} = T_{par} P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -d \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Transformation Matrix – Parallel Projection

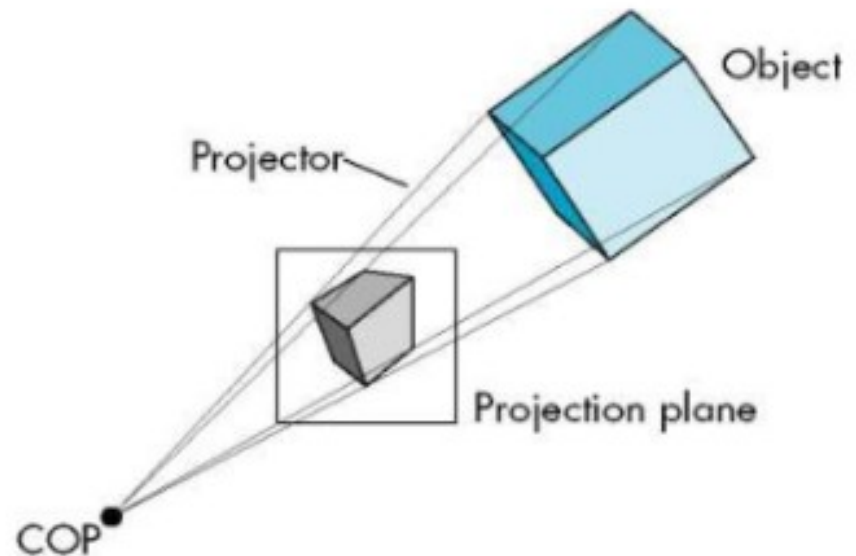
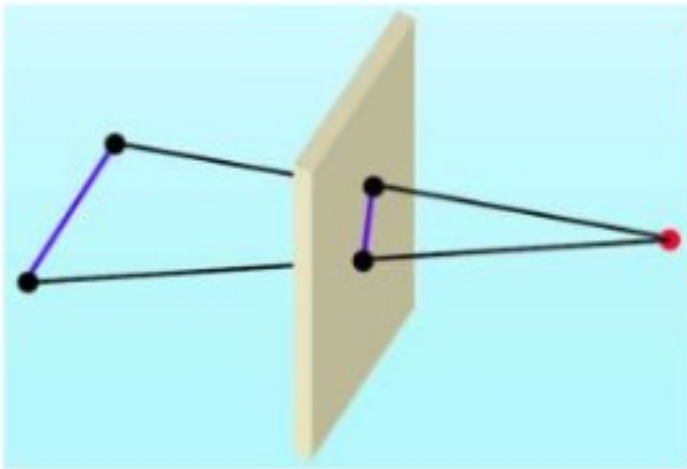
- Note multiplication is performed in homogeneous coordinate system.
- Therefore, real coordinates of P'

$$x' = \frac{x''}{w}, y' = \frac{y''}{w}, \text{ and } z' = \frac{z''}{w}$$



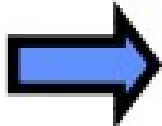
Perspective Projection

- Projectors are extended from object to plane of projection.
- Projectors are not parallel
- Projectors converge to single point called center of projection.
- Does not preserve relative proportion.
- Gives realistic view of an object and 3D impression.



Perspective Projection

- In a perspective projection, the center of projection is at a finite distance from the viewing plane.
- Parallel lines that are not parallel to the viewing plane, converge to a *vanishing point*.

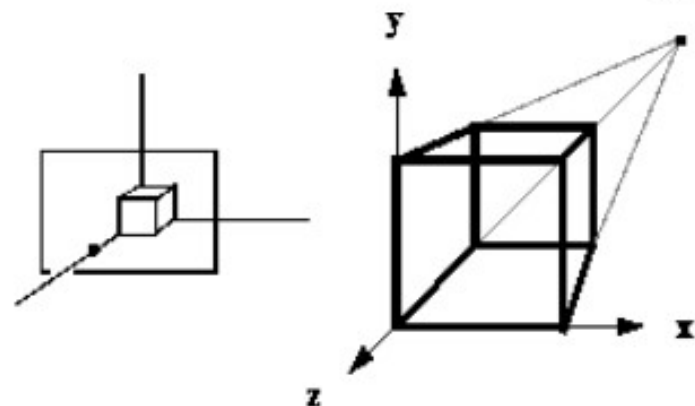
 A vanishing point is the projection of a point at infinity.



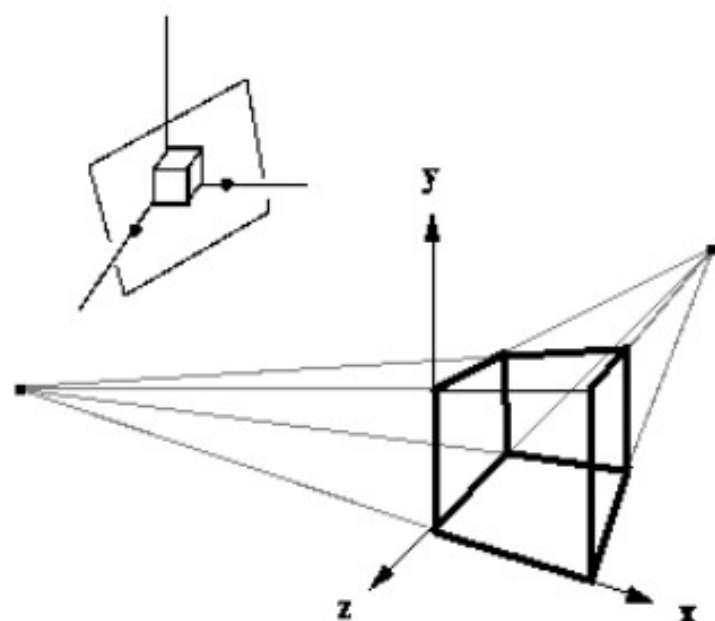
Perspective Projection

- There are infinite vanishing points.
- Vanishing points at axis are principal vanishing points.
- Vanishing points makes the projected object to appear 3D.

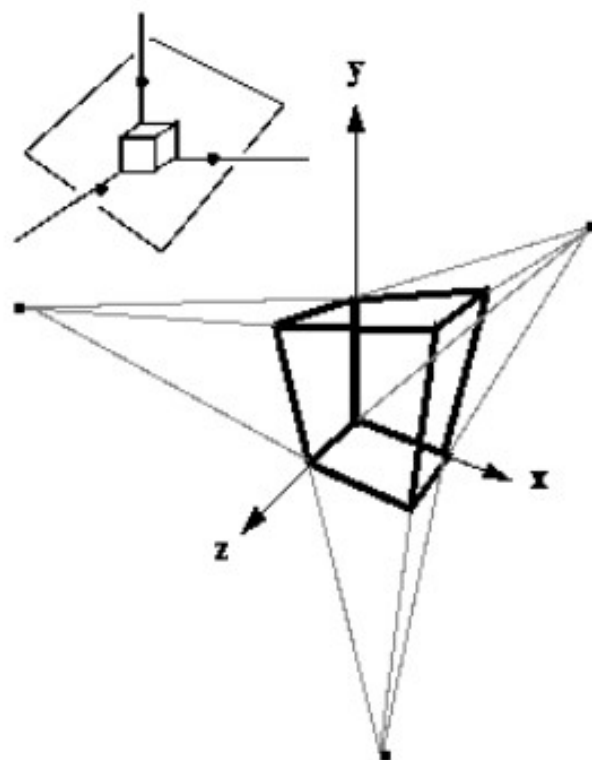
Vanishing Points



One Point Perspective
(z-axis vanishing point)



Two Point Perspective
z, and x-axis vanishing points



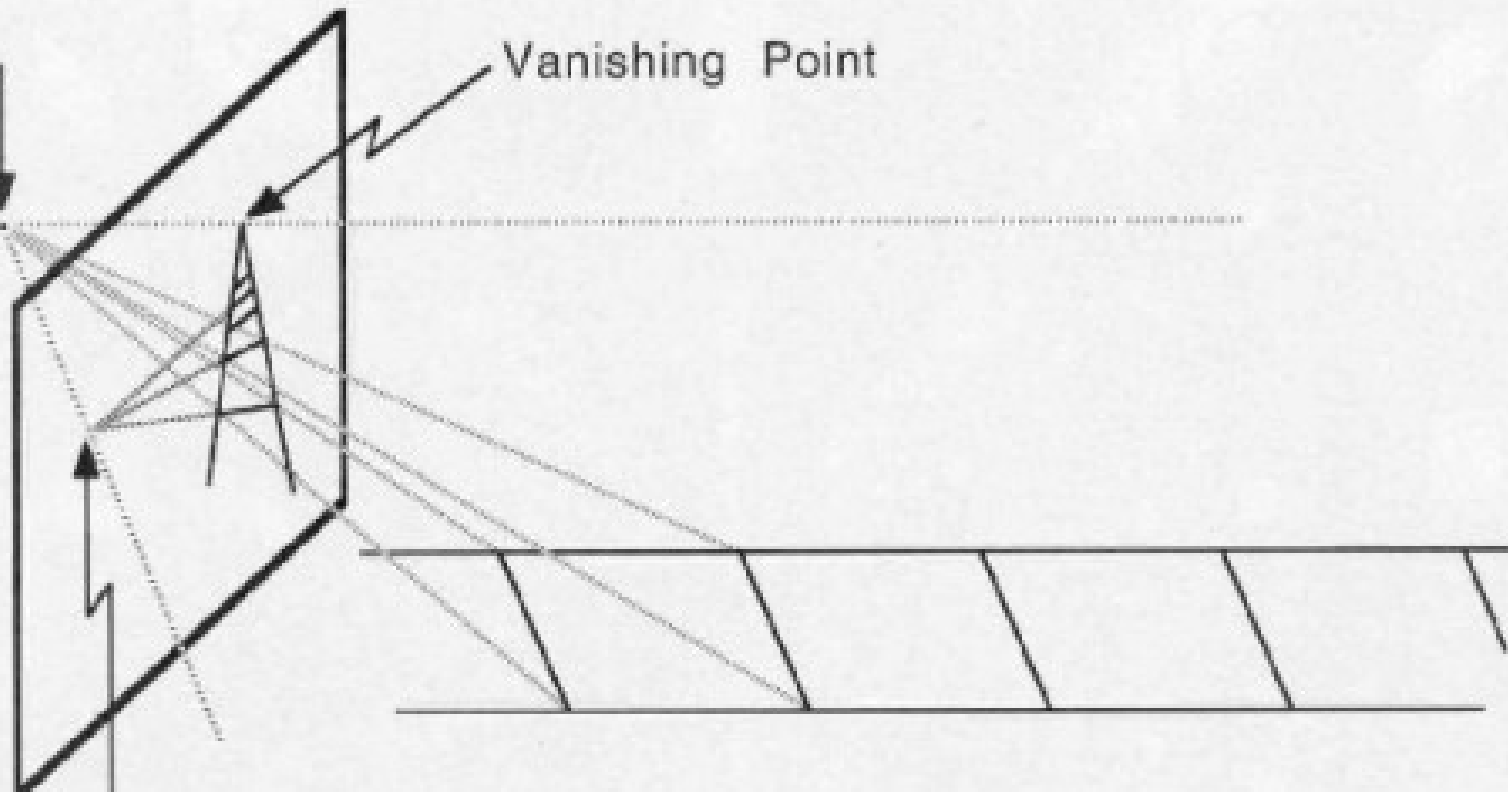
Three Point Perspective
(z, x, and y-axis
vanishing points)

Center of
Projection

Projection
Plane

Vanishing Point

Vanishing Point



Types of perspective drawing

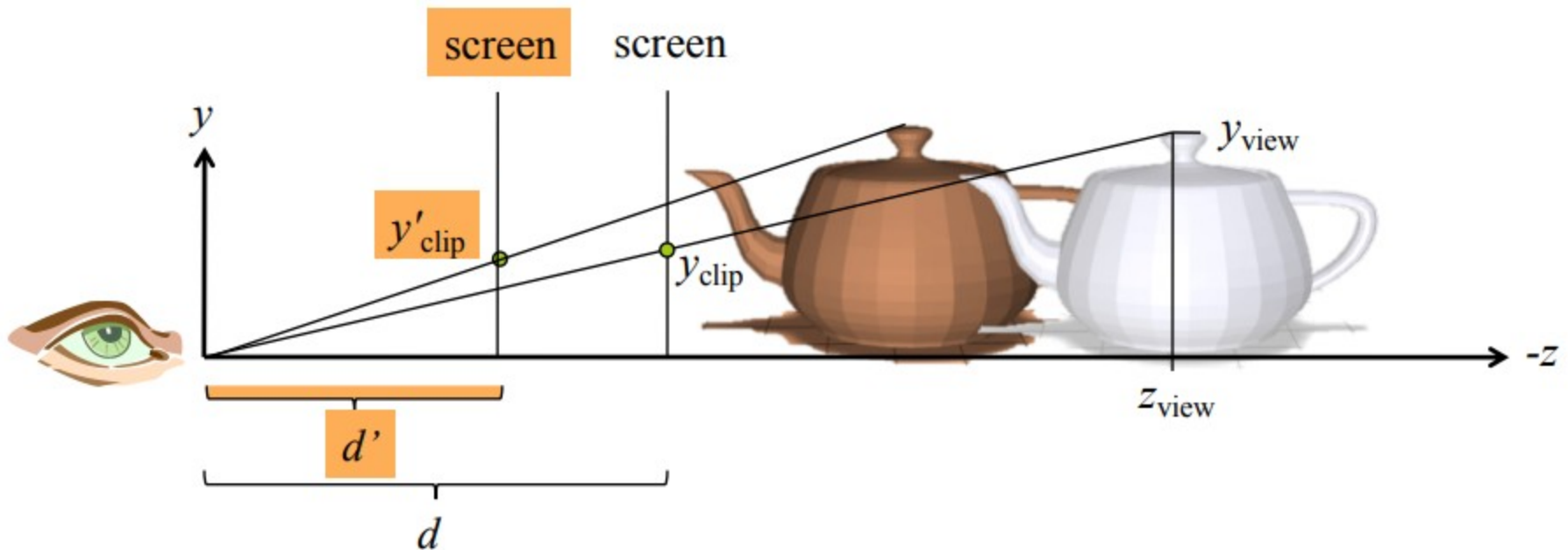
Perspective drawings are often classified by the number of principal vanishing points.

- ◆ One-point perspective — simplest to draw
- ◆ Two-point perspective — gives better impression of depth
- ◆ Three-point perspective — most difficult to draw

All three types are equally simple with computer graphics.

Foreshortening Effect

- Objects closer to view plane appears bigger.



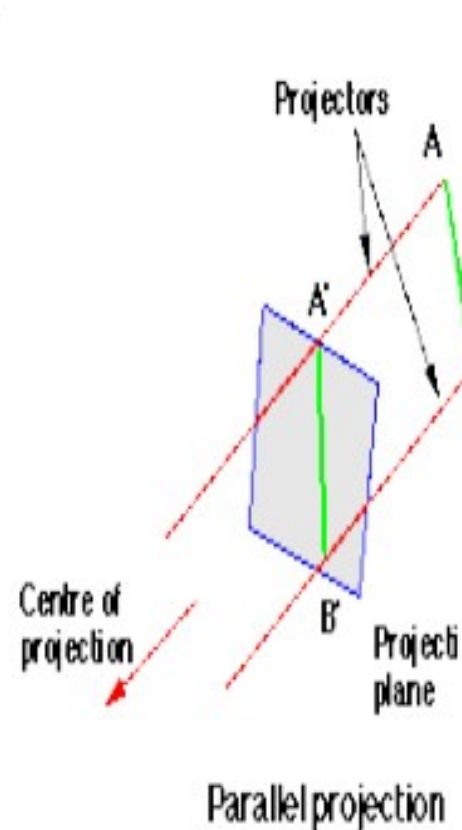
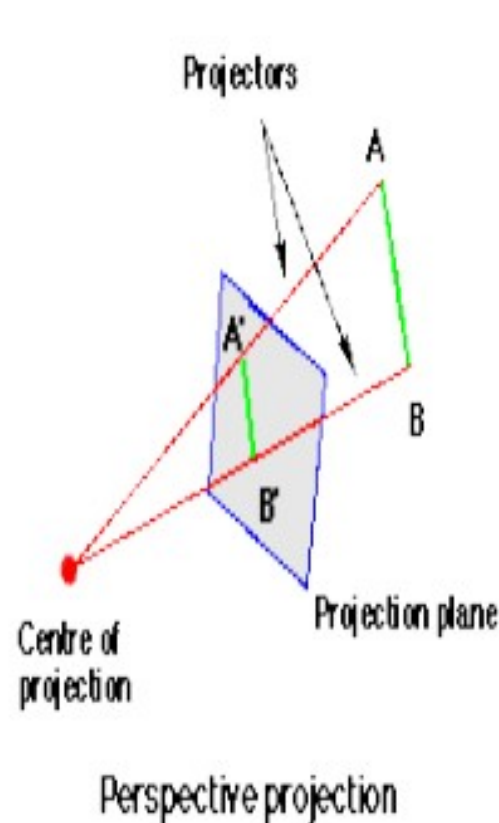
Comparison between Parallel and Perspective

- **Parallel Projections:**

- The center of projection is at infinity.
- The projectors are parallel to each other.

- **Perspective Projections:**

- The center of projection is a finite point.
- The projectors intersect at the center of projection.



Perspective vs. parallel projections

Perspective projections pros and cons:

- + Size varies inversely with distance - looks realistic
- Distance and angles are not (in general) preserved
- Parallel lines do not (in general) remain parallel

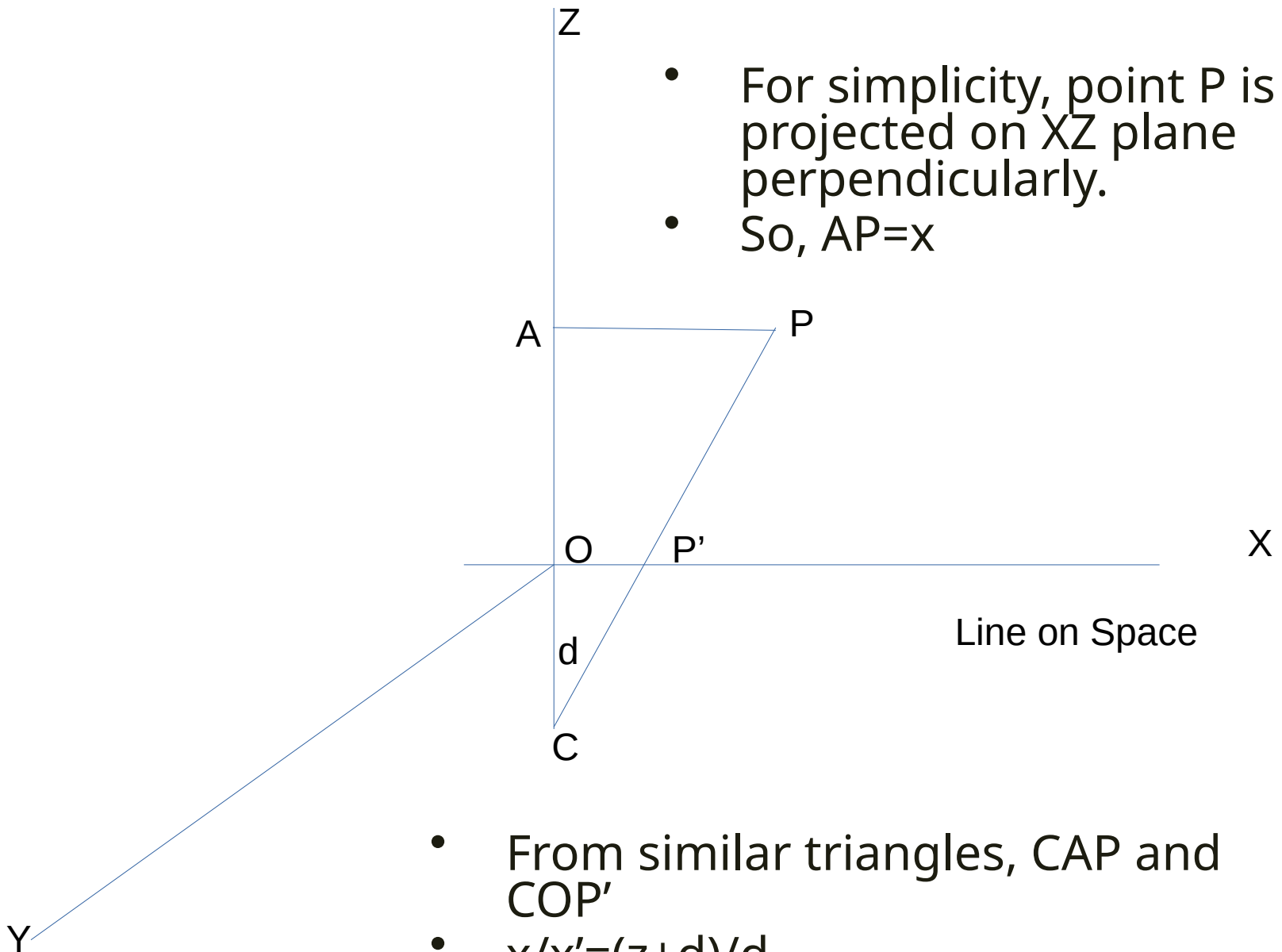
Parallel projection pros and cons:

- Less realistic looking
- + Good for exact measurements
- + Parallel lines remain parallel
- Angles not (in general) preserved

Perspective Projection Matrix

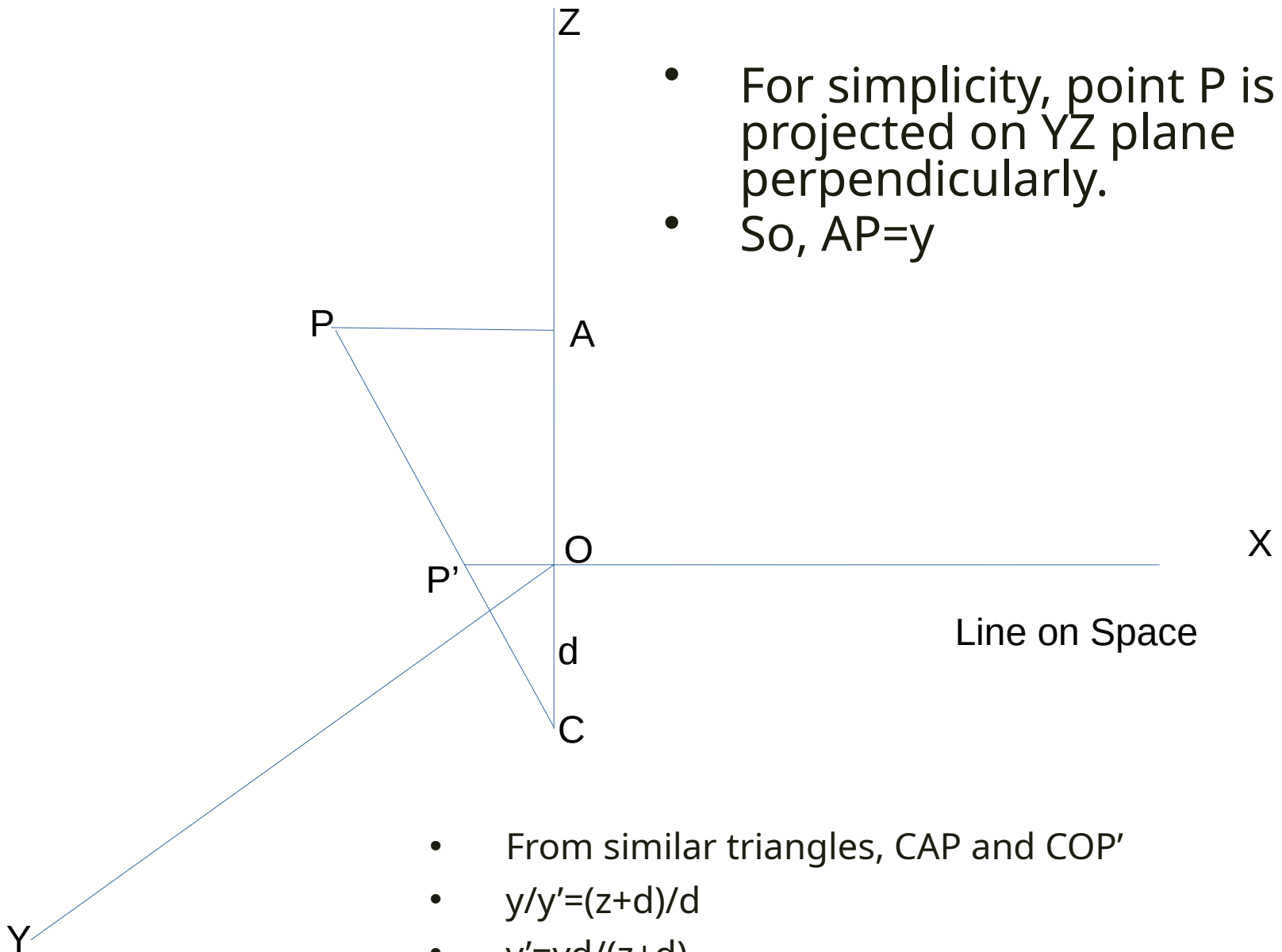
- For perspective projection matrix, let us assume that view plane is XY plane at origin
- And COP is at negative z axis at d distance from view plane.
- Line AP is projected on view plane as OP by extending lines from A and P towards COP.
- $P(x,y,z)$ and $A(0,0,z)$ as shown in the figure.

- For simplicity, point P is projected on XZ plane perpendicularly.
- So, $AP = x$



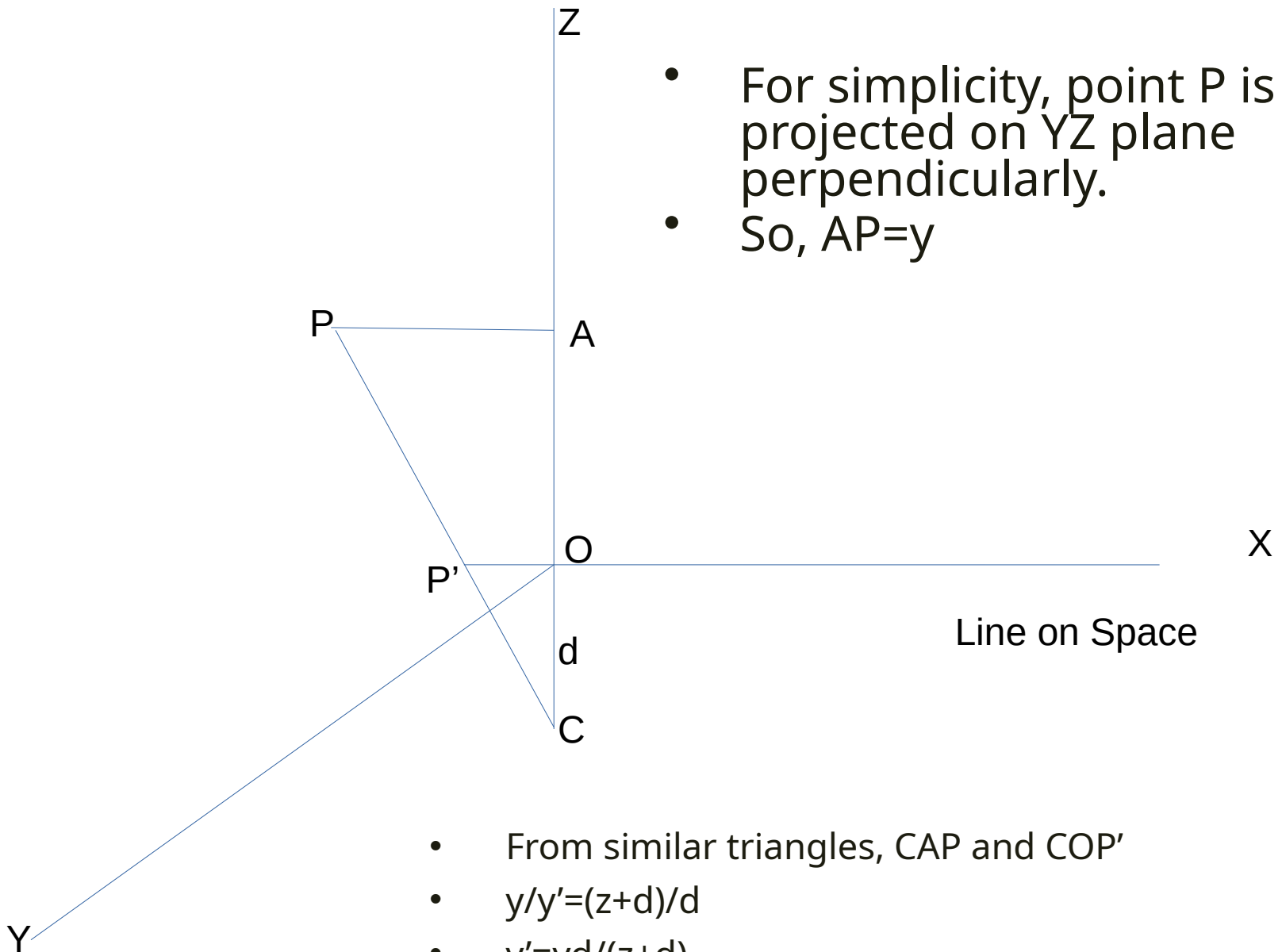
- From similar triangles, CAP and COP'
- $x/x' = (z+d)/d$
- $x' = xd/(z+d)$

- For simplicity, point P is projected on YZ plane perpendicularly.
- So, $AP=y$



- From similar triangles, CAP and COP'
- $y/y' = (z+d)/d$
- $y' = yd/(z+d)$
- $z' = 0$

- For simplicity, point P is projected on YZ plane perpendicularly.
- So, $AP=y$



- From similar triangles, CAP and COP'
- $y/y' = (z+d)/d$
- $y' = yd/(z+d)$
- $z' = 0$

Prospective in homogenous

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{d \cdot x}{z+d} \\ \frac{d \cdot y}{z+d} \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \frac{1}{z+d} \begin{bmatrix} d \cdot x \\ d \cdot y \\ 0 \\ z+d \end{bmatrix} = \frac{1}{z+d} \begin{bmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & d \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$